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Official Course
Outline

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Course Outline
(Syllabus)

3. Proposal: 3-Minute Talk (3MT)



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Assignment

Due October 15 at 5:00 AM

Please see the attached manual and sample submission. The sample is **for illustration only and is not a complete or authoritative example**. Please follow the manual for the actual requirements.

A cinematic still from the movie Pulp Fiction. It shows Mia Wallace (Uma Thurman) in the passenger seat and Vincent Vega (John Travolta) in the driver's seat of a bright red sports car. They are both looking forward with serious expressions. The car's interior is black with a red checkered pattern on the dashboard. The background is dark, suggesting a nighttime setting with some blurred lights.

Don't be squared ..., Pulp Fiction, Quintine Tarantino, 1994

<https://www.youtube.com/watch?v=IY48IXSK4nk>

3

A Data Odyssey: Relational DBMS vs. GGG



Data Rep.
in
RDBMS

Real World Entity

Conceptual Level | Entity-Relationship Model Level

| Logical Level | Relational Model

| Physical Level | SQL

| Computable Entity



Data Rep.
in
GGG

Real World Entity

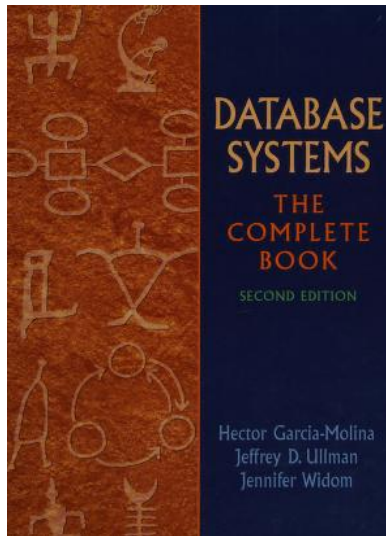
Conceptual Level | Web Ontology Language (OWL)

| Logical Level | RDF & Description Language

| Physical Level | SPARQL

| Computable Entity

Outline: W04-05



CH02 (2nd Ed.)

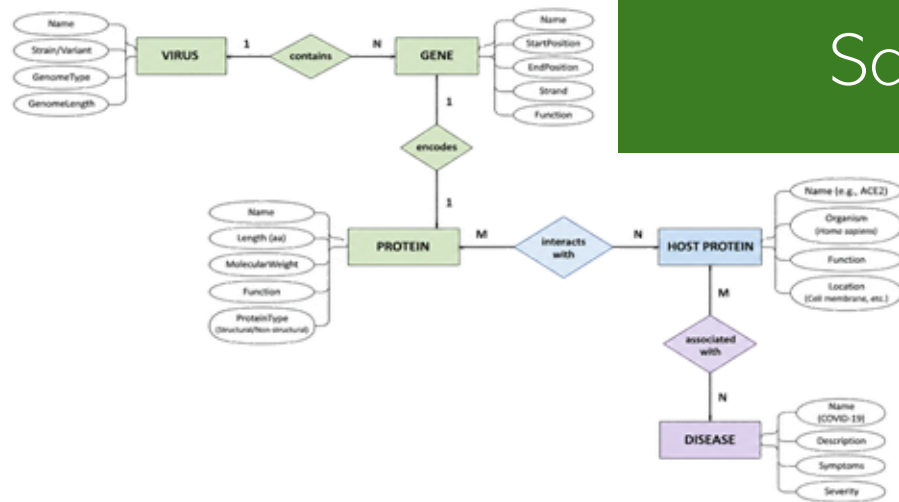
- I) Relational Modeling
- II) Entity to Relation
- III) Relationship to Relation
- IIII) IS-A to Relation

Hands-On

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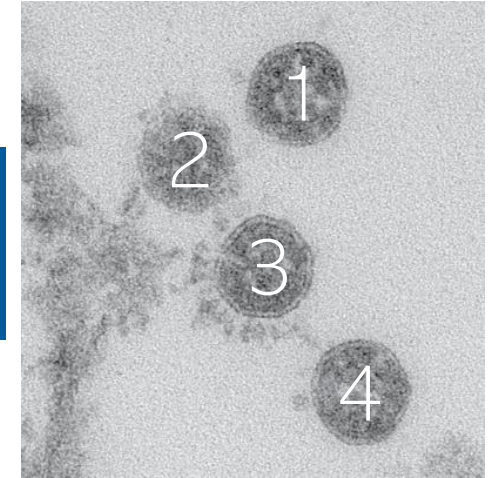
Abstractions: Logical Representation

How **instances** of entities, **values** of their attributes, their relationships should be represented based on a schema.



Schema

Real-World
Instances of Entity



Relational Model: Formal vs. Informal Terminology

Relation

Schema	<u>Title</u>	ReleaseDate	RunningTime	Genre
Tuple 1	2001: A Space Odyssey	English	1 42	SciFi
Tuple 2	Rosemary's Baby	English	1 36	Horror
Tuple 3	The Birds	English	1 19	Horror
...	...	...	...	

Relation vs. Table

Tuple vs. Row

Domain vs. Column

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Relational Model: Everything is Relation

Entity Set $\rightarrow$ Relation

Relationship Set $\rightarrow$ Relation

Data Modeling
in
RDBMS

Real World Entity

Conceptual Level | Entity Set & Relationship Set

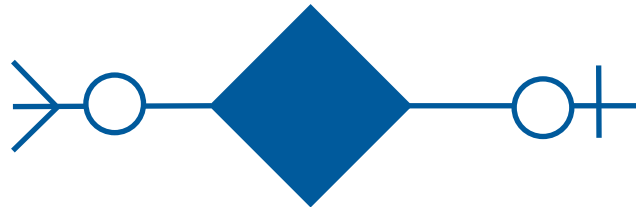
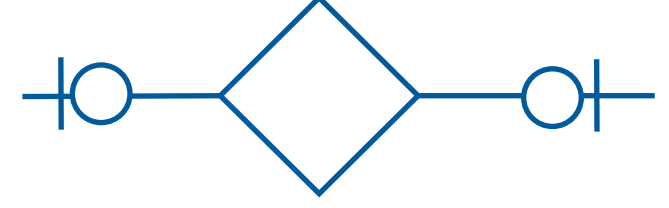
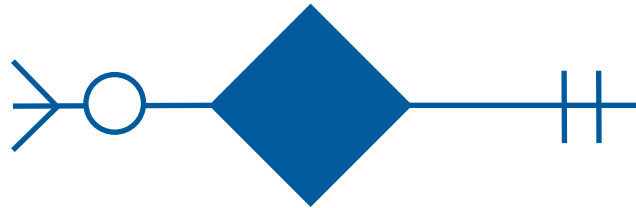
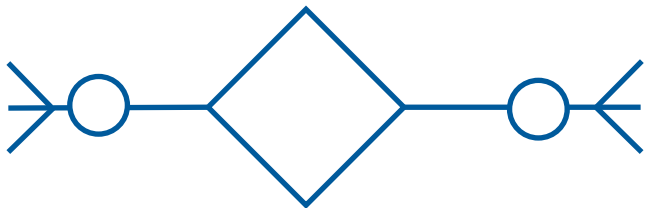
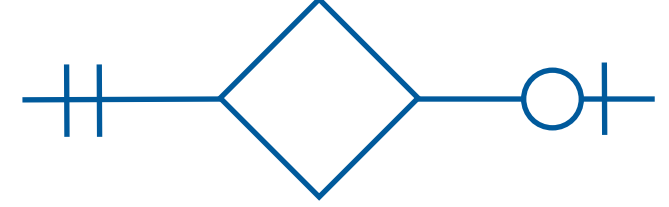
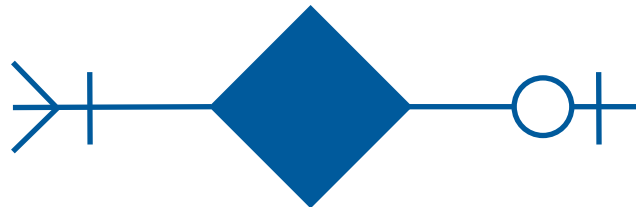
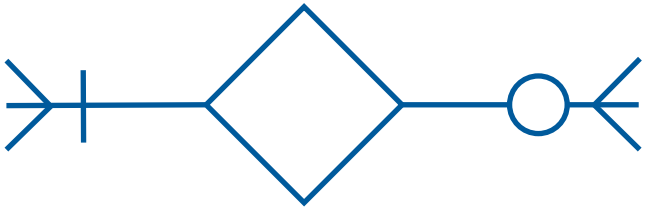
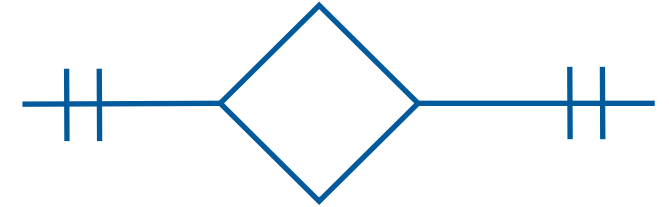
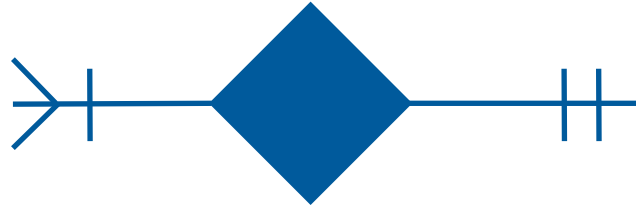
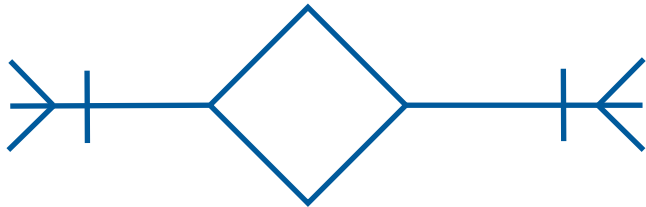
Conceptual Level | Logical Level | Relation

Conceptual Level | Logical Level | Physical Level | SQL

Conceptual Level | Logical Level | Computable Entity

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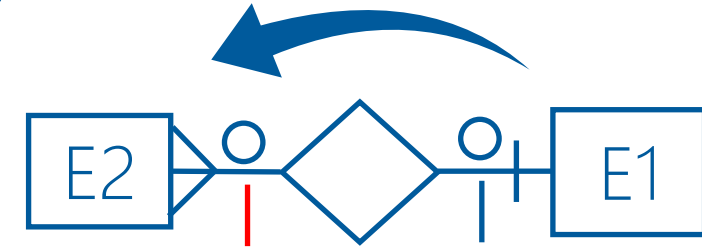
Relationship: Many-One (One-Many)



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Relationship: Many-One (One-Many)

Input: Many-One relationship between E2 and E1, i.e.,
Output: Relations R1 for E1 and R2 for E2.

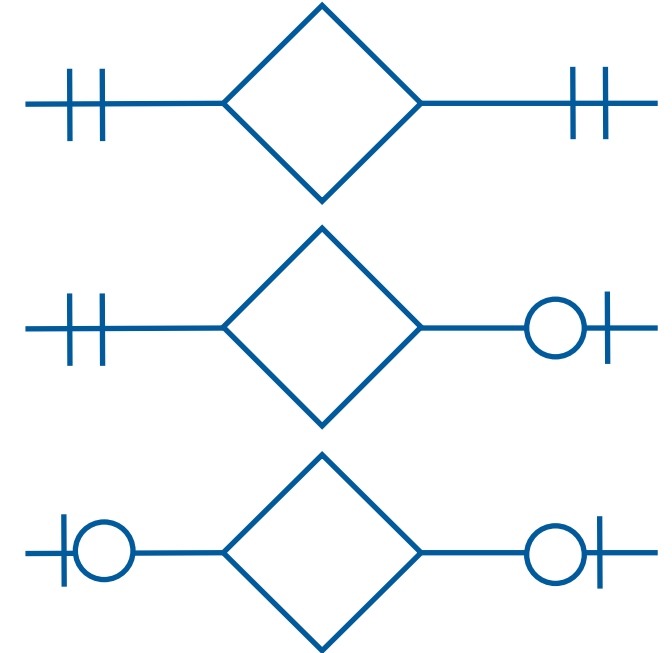
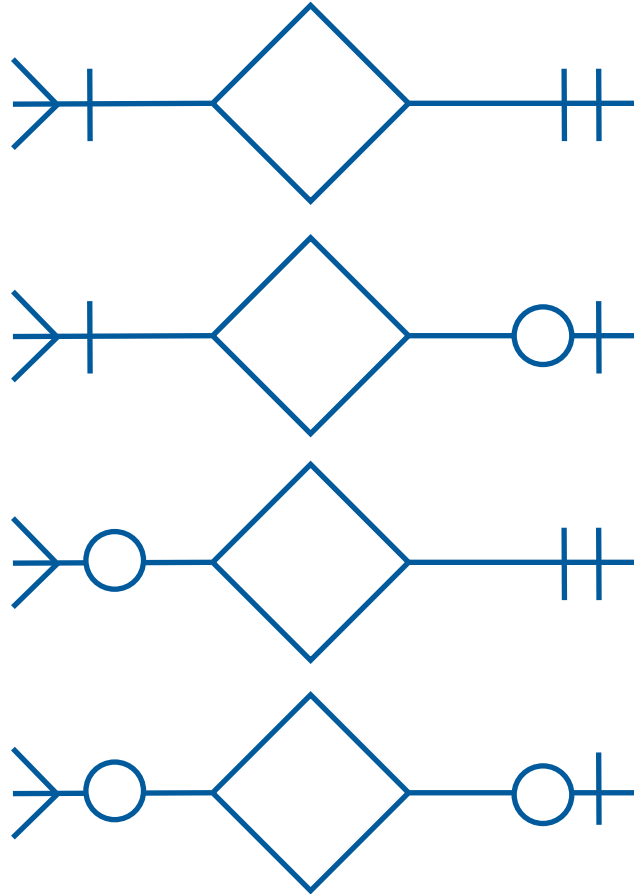
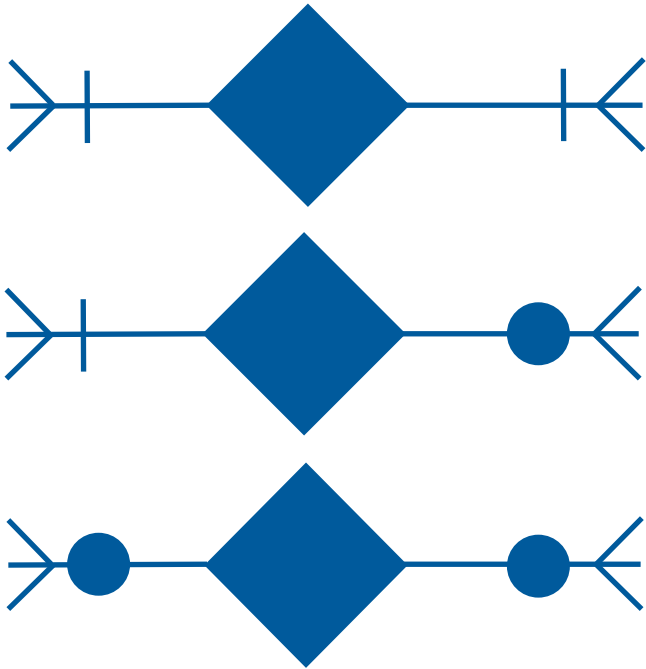


1. For E1, create relation R1 with the same attributes and keys as in E1
2. For E2, create relation R2 with the same attributes and keys as in E2
3. **[Foreign Key Set]** Add key set of E1 to R2
4. Add attributes of relationship set to R2
5. If E2 ordinality is optional then make foreign key set optional
else make foreign key set mandatory

Herein, we do not care about E1's ordinality.

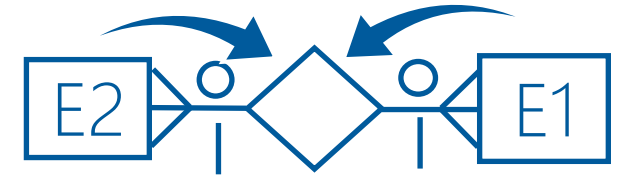
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Outline: W05A



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Relationship to Relations: Many-Many



Input: Many-Many relationship btw. E2 and E1, i.e.,

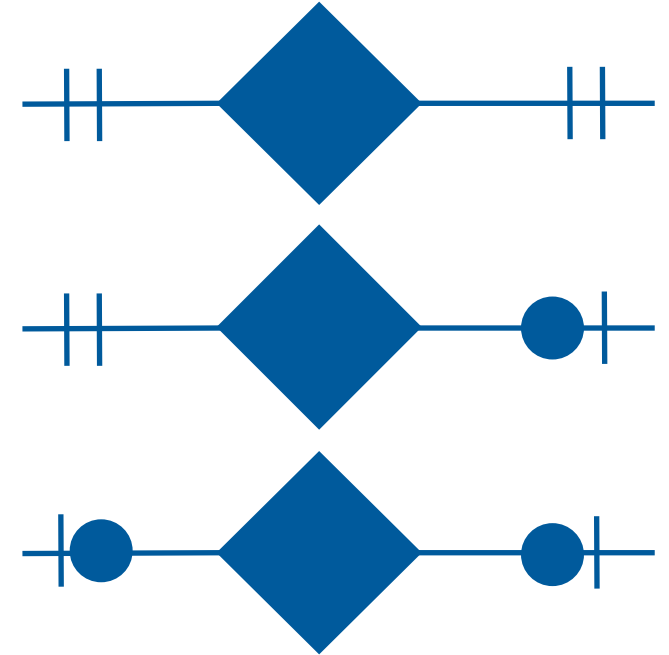
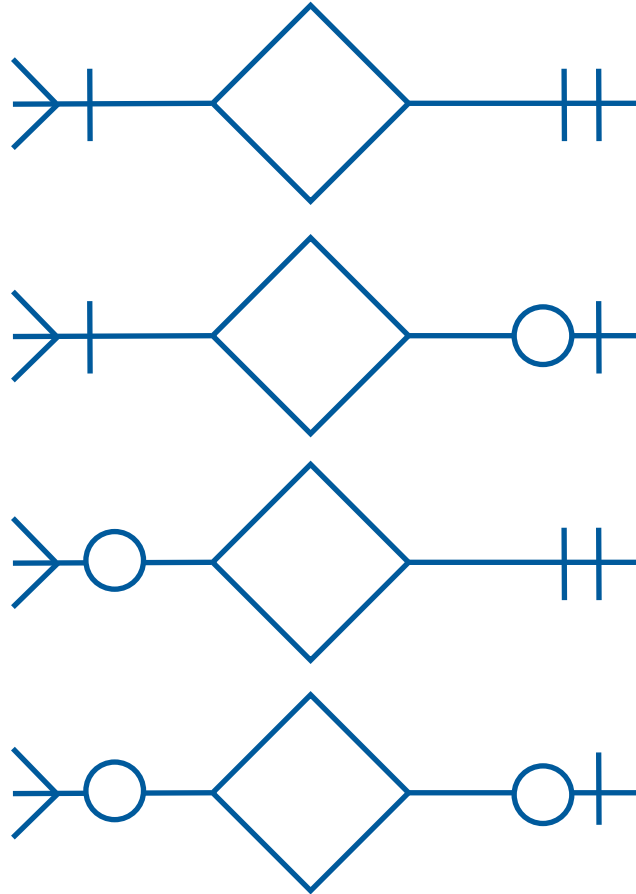
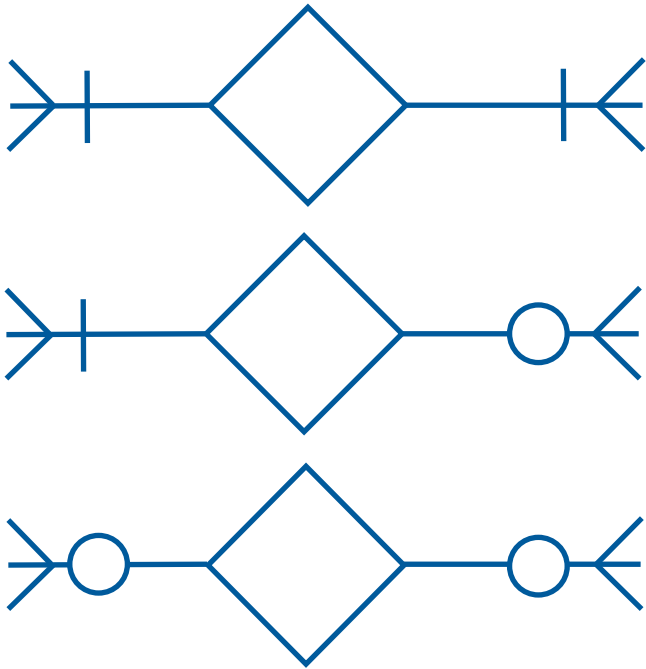
Output: Relations R1 for E1, R2 for E2, and R3 for relationship.

- 1) For E1, create relation R1 with the same attributes and keys as in E1
- 2) For E2, create relation R2 with the same attributes and keys as in E2
- 3) For many-many relationship set, **create new relation R3**
- 4) **[Foreign Key Set FK_1]** Add key set of E1 to R3
- 5) **[Foreign Key Set FK_2]** Add key set of E2 to R3
- 6) Add attributes of relationship set to R3
- 7) **[Primary Key Set]** Create key set for R3 from foreign keys (step 4 & 5) $PK = FK_1 \cup FK_2$

Herein, we do not care about ordinality! We fix it later.

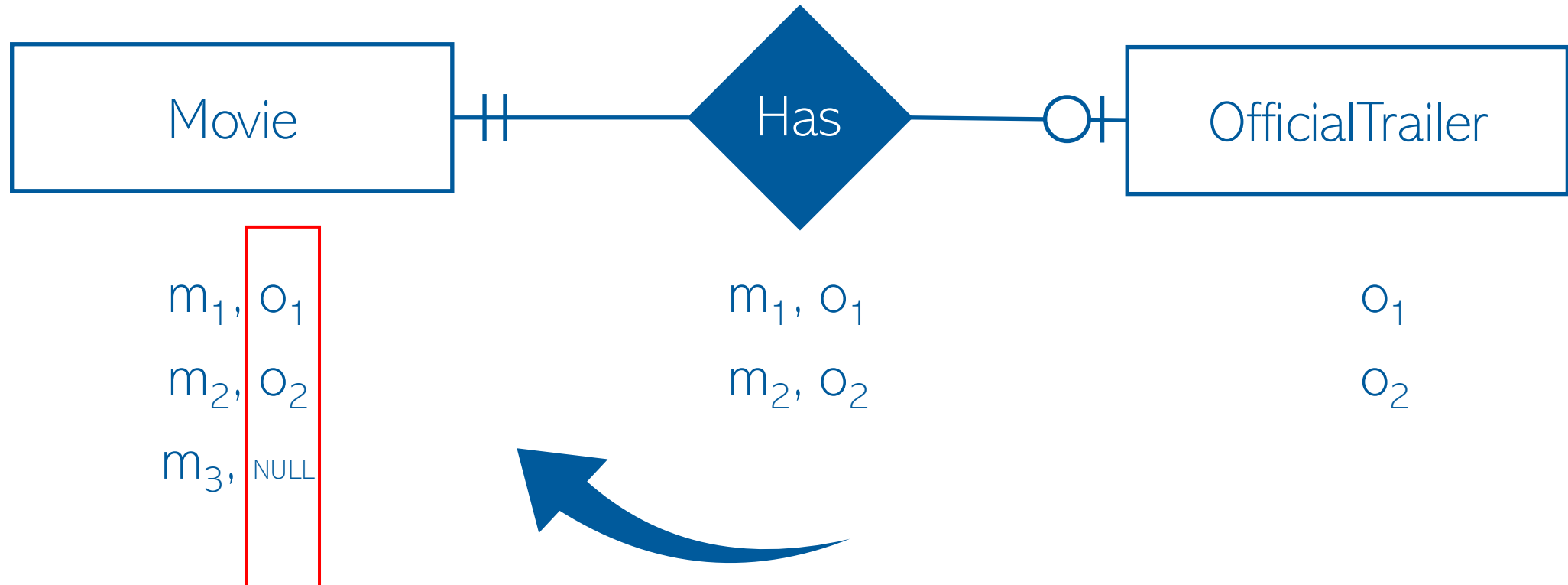
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Outline: W05A



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Relationship to Relations: One-One (Approach A)



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Relationship to Relations: For Any Relations

Candidate Key (CK):

A key set whose attributes can accept no value (NULL).

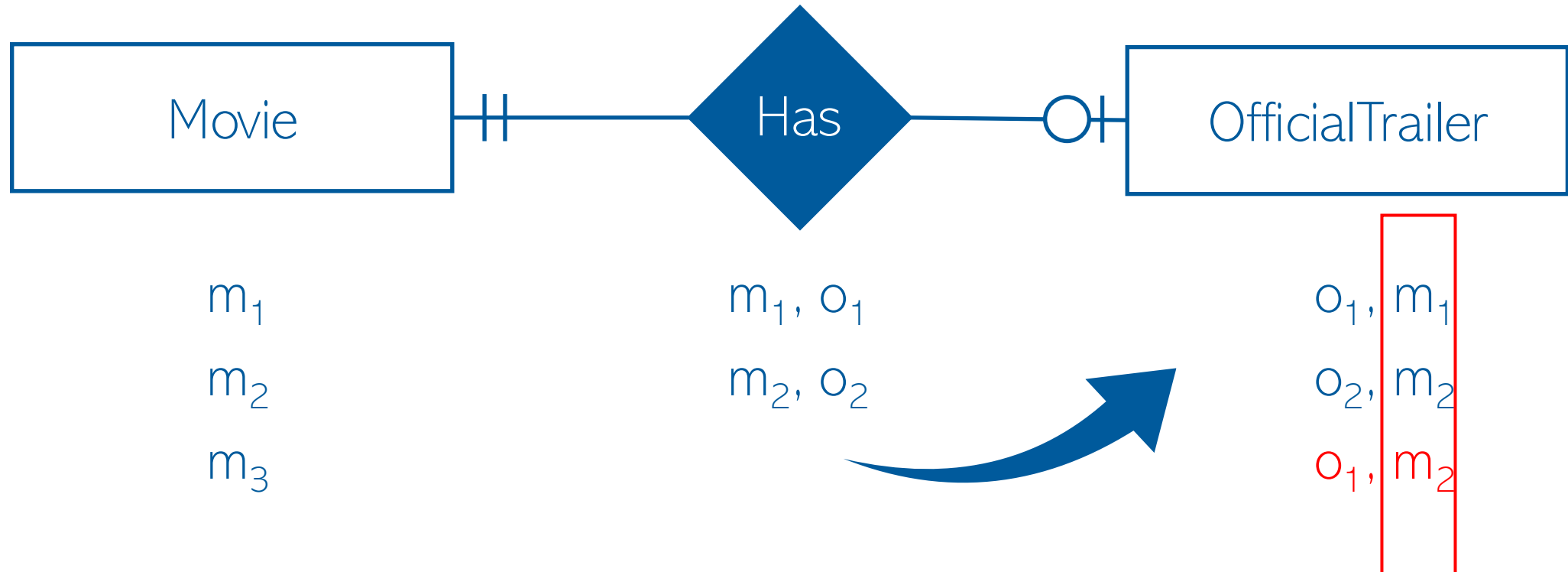
In Primary Key, all its attributes must have value, but CK's attributes could be optional.

So, relational model allows multiple key sets for a relation (table). It is *not* called Secondary Key set ... It's called Candidate Key (CK).

ER does not have it.

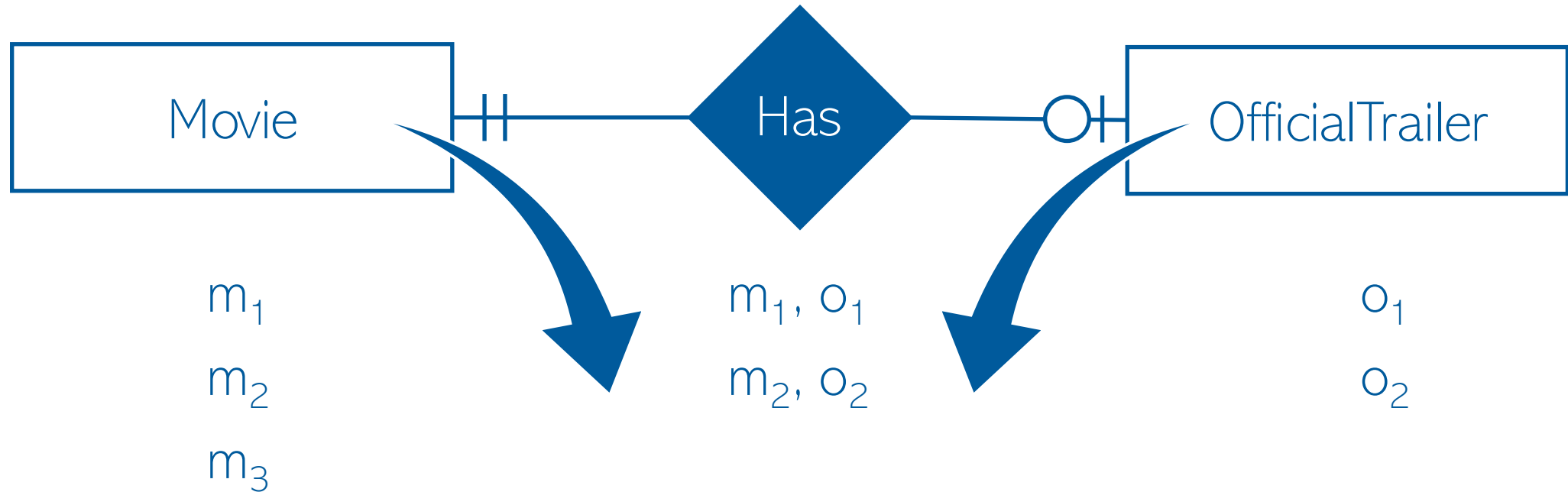
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Relationship to Relations: One-One (Approach B)



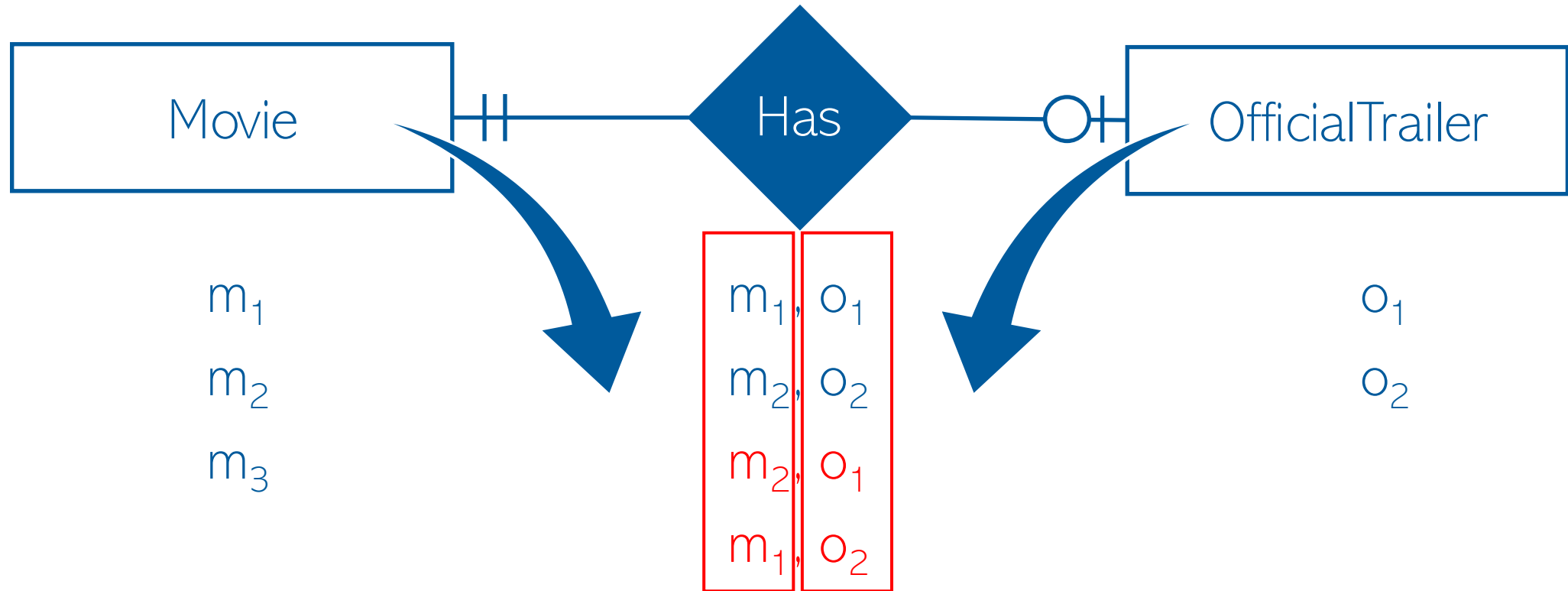
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Relationship to Relations: One-One (Approach C)



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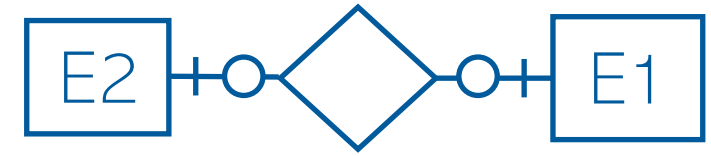
Relationship to Relations: One-One (Approach C)



Each domain (column) must be unique ...

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Relationship to Relations: One-One



Input: One-One relationship btw. E2 and E1, i.e.,

Output: Relations R1 for E1, R2 for E2, and/or R3 for relationship.

- 1) For E1, create relation R1 with the same attributes and keys as in E1
- 2) For E2, create relation R2 with the same attributes and keys as in E2
- 3) Do only one of the followings
 - a) [Foreign & Candidate Key set FK, CK] Add key set of E1 to R2 & make them candidate key of R2
If E1's ordinality is mandatory, make CK mandatory
 - b) [Foreign & Candidate Key Set FK, CK] Add key set of E2 to R1 & make them candidate key of R1
If E2's ordinality is mandatory, make CK mandatory
 - c) Create new relation R3
 - I) [Foreign & Primary Key set FK, PK] Add key set of E1 to R3 & make them primary key of R3
 - II) [Foreign & Candidate Key set FK, CK] Add key set of E2 to R3 & make them candidate key of R3 and mandatory



Divine Intervention ..., Pulp Fiction, Quintine Tarantino, 1994
<https://www.youtube.com/watch?v=l2oaQ3Qv7pQ>

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Surrogate Key vs. Natural (Business) Key Sets

Sometimes Primary Key set gets large,

E.g., Movie's PK={Title, ReleaseYear, ReleaseMonth, ReleaseDay}

E.g., Actor's PK={FirstName, LastName, BirthDate, BirthPlace}

It replicates when becomes FK to other relations.

Not convenient!

Surrogate Key vs. Natural (Business) Key Sets

- One single domain (attribute) key (not a set)
- System-generated, e.g., AUTONUMBER, i.e., 1,2,3,....,
- The value has no meaning in real-world!
- Just for data engineering convenience.
- Usually named Id, ID, RowID, RowId, TupleId, ...

Old natural (business) primary key becomes mandatory candidate key.

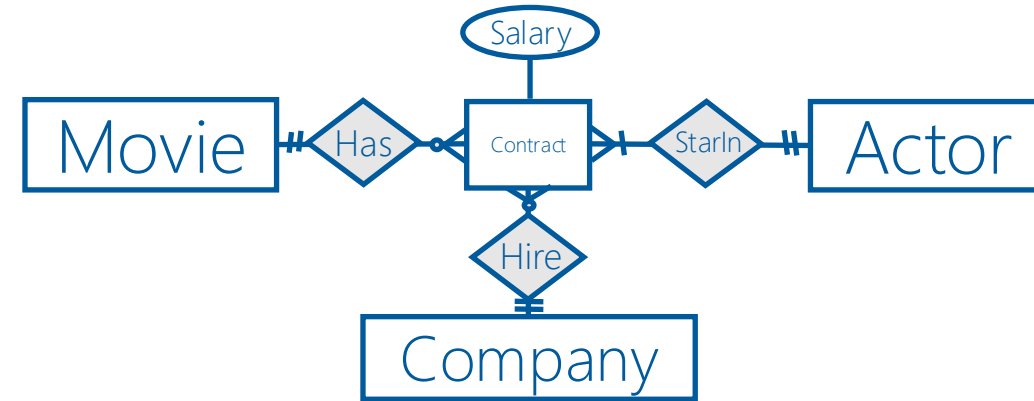
Old natural (business) primary key becomes mandatory candidate key.

Old natural (business) primary key becomes mandatory candidate key.

Old natural (business) primary key becomes mandatory candidate key.

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Surrogate Key vs. Natural (Business) Key Sets



R1: Company(Id, Name, Address, Phone, ...)

CK={Name} Mandatory

R2: Actor(Id, FirstName, LastName, BirthDate, BirthPlace, ...)

CK={FirstName, LastName, BirthDate, BirthPlace} Mandatory

R3: Movie(Id, Title, ReleaseYear, ReleaseMonth, ReleaseDay, RunningTime)

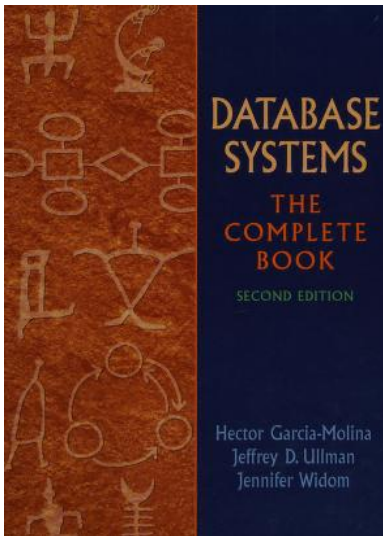
CK={Title, ReleaseYear, ReleaseMonth, ReleaseDay} Mandatory

R4: Contract(Id, Salary, Company.Id, Actor.Id, Movie.Id)

FK₁={Company.Id}, FK₂={Actor.Id}, FK₃={Movie.Id}

CK=FK₁ U FK₂ U FK₃ Mandatory

Outline: W05A



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

III) Relationship to Relation (One-One)

III) IS-A to Relation

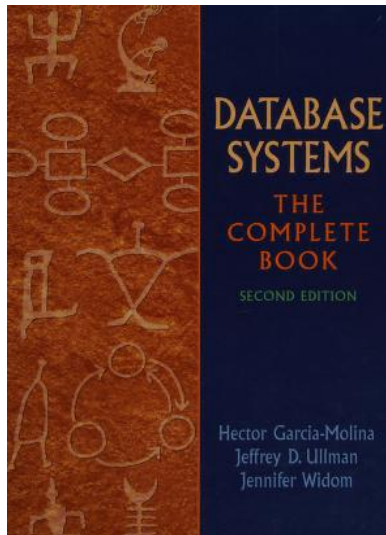
Hands-On



Trunk POV..., Pulp Fiction, Quintine Tarantino, 1994
<https://www.youtube.com/watch?v=edFrWONapIE>

ISA to Relation

Outline: W05A



CH02 (2nd Ed.)

I) Relational Modeling

II) Entity to Relation

III) Relationship to Relation (One-One)

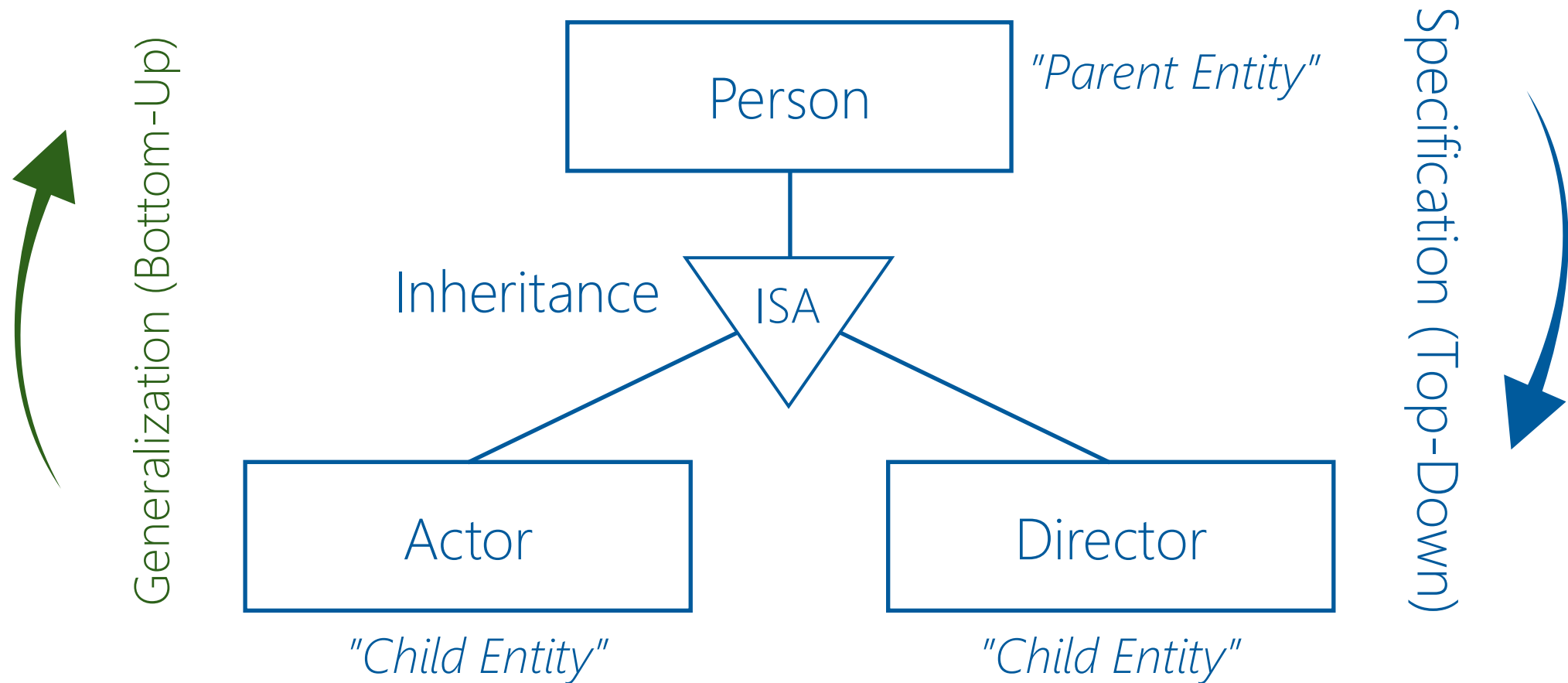
III) IS-A to Relation

Hands-On

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IS-A (Inheritance, Generalization $\leftrightarrow$ Specification)

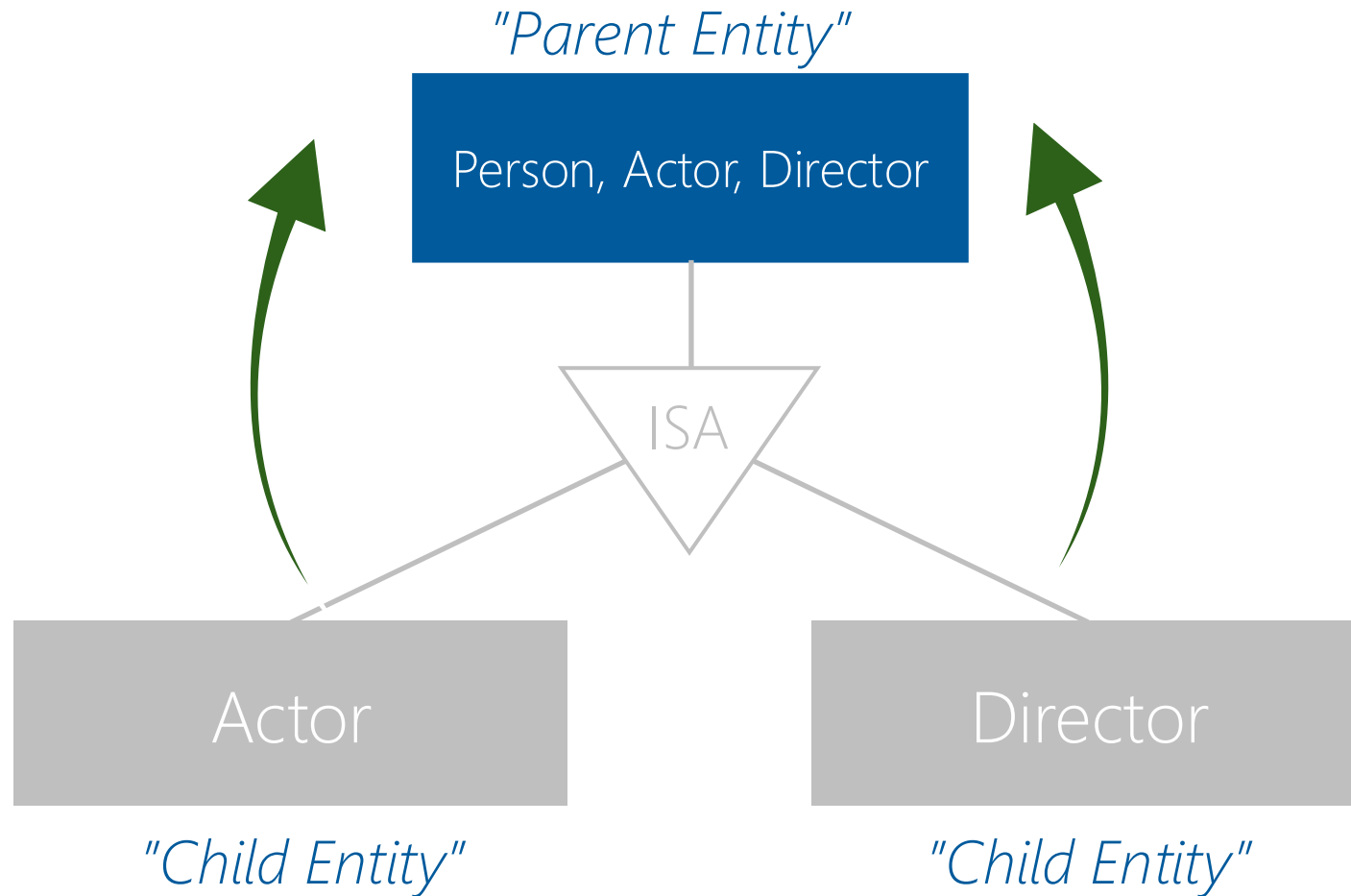
Both are person, so they have person's attributes like name, age, ...



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IS-A to Relation (Approach I)

All goes to parent as if no IS-A ...



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IS-A to Relation (Approach I)

All goes to parent as if no IS-A ...

"Parent Entity"

Person, Actor, Director

R1: Person(Id, FirstName, LastName, BirthDate, BirthPlace, ...

PersonType,

ActorBestLine, ActorMovieCount,

DirectorBestMovie, DirectorMovieCount)

CK={FirstName, LastName, BirthDate, BirthPlace}

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IS-A to Relation (Approach I)

All goes to parent as if no IS-A ...

Id	FirstName	LastName	BirthDate	BirthPlace	PersonType	ActorBestLine	ActorMovieCount	DirectorBestMovie	DirectorMovieCount
1	Stanley	Kubrick	Jul. 26, 1928	USA	Director	NULL	NULL	102	13
2	Alfred	Hitchcock	Aug. 13, 1899	England	Director	NULL	NULL	203	47
3	John	Travolta	Feb. 18, 1954	USA	Actor	You ...	61	NULL	NULL
4	Samuel	Jackson	Dec. 21, 1948	USA	Actor	Say 'w...	125	NULL	NULL
5	Uma	Thurman	Apr. 29, 1970	USA	Actor	I believe ..	51	NULL	NULL
6	Hossein	Fani	Mar. 11, 1844	Iran	NULL	NULL	NULL	NULL	NULL
7	Clint	Eastwood	May 31, 1930	USA	ActorDirector	A good ..	69	803	35

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IS-A to Relation (Approach I)

All goes to parent as if no IS-A ... Partial

Id	FirstName	LastName	BirthDate	BirthPlace	PersonType	ActorBestLine	ActorMovieCount	DirectorBestMovie	DirectorMovieCount
1	Stanley	Kubrick	Jul. 26, 1928	USA	Director	NULL	NULL	102	13
2	Alfred	Hitchcock	Aug. 13, 1899	England	Director	NULL	NULL	203	47
3	John	Travolta	Feb. 18, 1954	USA	Actor	You ...	61	NULL	NULL
4	Samuel	Jackson	Dec. 21, 1948	USA	Actor	Say 'w...	125	NULL	NULL
5	Uma	Thurman	Apr. 29, 1970	USA	Actor	I believe ..	51	NULL	NULL
6	Hossein	Fani	Mar. 11, 1844	Iran	NULL	NULL	NULL	NULL	NULL
7	Clint	Eastwood	May 31, 1930	USA	ActorDirector	A good ..	69	803	35

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IS-A to Relation (Approach I)

All goes to parent as if no IS-A ... Overlapping

Id	FirstName	LastName	BirthDate	BirthPlace	PersonType	ActorBestLine	ActorMovieCount	DirectorBestMovie	DirectorMovieCount
1	Stanley	Kubrick	Jul. 26, 1928	USA	Director	NULL	NULL	102	13
2	Alfred	Hitchcock	Aug. 13, 1899	England	Director	NULL	NULL	203	47
3	John	Travolta	Feb. 18, 1954	USA	Actor	You ...	61	NULL	NULL
4	Samuel	Jackson	Dec. 21, 1948	USA	Actor	Say 'w...	125	NULL	NULL
5	Uma	Thurman	Apr. 29, 1970	USA	Actor	I believe ..	51	NULL	NULL
6	Hossein	Fani	Mar. 11, 1844	Iran	NULL	NULL	NULL	NULL	NULL
7	Clint	Eastwood	May 31, 1930	USA	ActorDirector	A good ..	69	803	35

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IS-A to Relation (Approach I)

All goes to parent as if no IS-A ... Can it be partial & disjoint? Yes.

Id	FirstName	LastName	BirthDate	BirthPlace	PersonType	ActorBestLine	ActorMovieCount	DirectorBestMovie	DirectorMovieCount
1	Stanley	Kubrick	Jul. 26, 1928	USA	Director	NULL	NULL	102	13
2	Alfred	Hitchcock	Aug. 13, 1899	England	Director	NULL	NULL	203	47
3	John	Travolta	Feb. 18, 1954	USA	Actor	You ...	61	NULL	NULL
4	Samuel	Jackson	Dec. 21, 1948	USA	Actor	Say 'w...	125	NULL	NULL
5	Uma	Thurman	Apr. 29, 1970	USA	Actor	I believe ..	51	NULL	NULL
6	Hossein	Fani	Mar. 11, 1844	Iran	NULL	NULL	NULL	NULL	NULL
7	Clint	Eastwood	May 31, 1930	USA	ActorDirector	... good ..	69	803	35

Partial: Make PersonType optional

Disjoint: If there is a value, it must be from domain {Director, Actor}.

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IS-A to Relation (Approach I)

All goes to parent as if no IS-A ... Can it be total & disjoint? Yes.

Id	FirstName	LastName	BirthDate	BirthPlace	PersonType	ActorBestLine	ActorMovieCount	DirectorBestMovie	DirectorMovieCount
1	Stanley	Kubrick	Jul. 26, 1928	USA	Director	NULL	NULL	102	13
2	Alfred	Hitchcock	Aug. 13, 1899	England	Director	NULL	NULL	203	47
3	John	Travolta	Feb. 18, 1954	USA	Actor	You ...	61	NULL	NULL
4	Samuel	Jackson	Dec. 21, 1948	USA	Actor	Say 'w...	125	NULL	NULL
5	Uma	Thurman	Apr. 29, 1970	USA	Actor	I believe ..	51	NULL	NULL
6	Hossein	Fani	Mar. 11, 1844	Iran	NULL	NULL	NULL	NULL	NULL
7	Clint	Eastwood	May 31, 1930	USA	ActorDirector	... good ..	69	803	35

Partial: Make PersonType mandatory

Disjoint: The value must be from domain {Director, Actor}.

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IS-A to Relation (Approach I)

All goes to parent as if no IS-A ... Can it be total & overlapping? Yes.

Id	FirstName	LastName	BirthDate	BirthPlace	PersonType	ActorBestLine	ActorMovieCount	DirectorBestMovie	DirectorMovieCount
1	Stanley	Kubrick	Jul. 26, 1928	USA	Director	NULL	NULL	102	13
2	Alfred	Hitchcock	Aug. 13, 1899	England	Director	NULL	NULL	203	47
3	John	Travolta	Feb. 18, 1954	USA	Actor	You ...	61	NULL	NULL
4	Samuel	Jackson	Dec. 21, 1948	USA	Actor	Say 'w...	125	NULL	NULL
5	Uma	Thurman	Apr. 29, 1970	USA	Actor	I believe ..	51	NULL	NULL
6	Hossein	Fani	Mar. 11, 1844	Iran	NULL	NULL	NULL	NULL	NULL
7	Clint	Eastwood	May 31, 1930	USA	ActorDirector	... good ..	69	803	35

Partial: Make PersonType mandatory

Disjoint: The value must be from domain {Director, Actor, ActorDirector}.

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IS-A to Relation (Approach I)

All goes to parent as if no IS-A ...

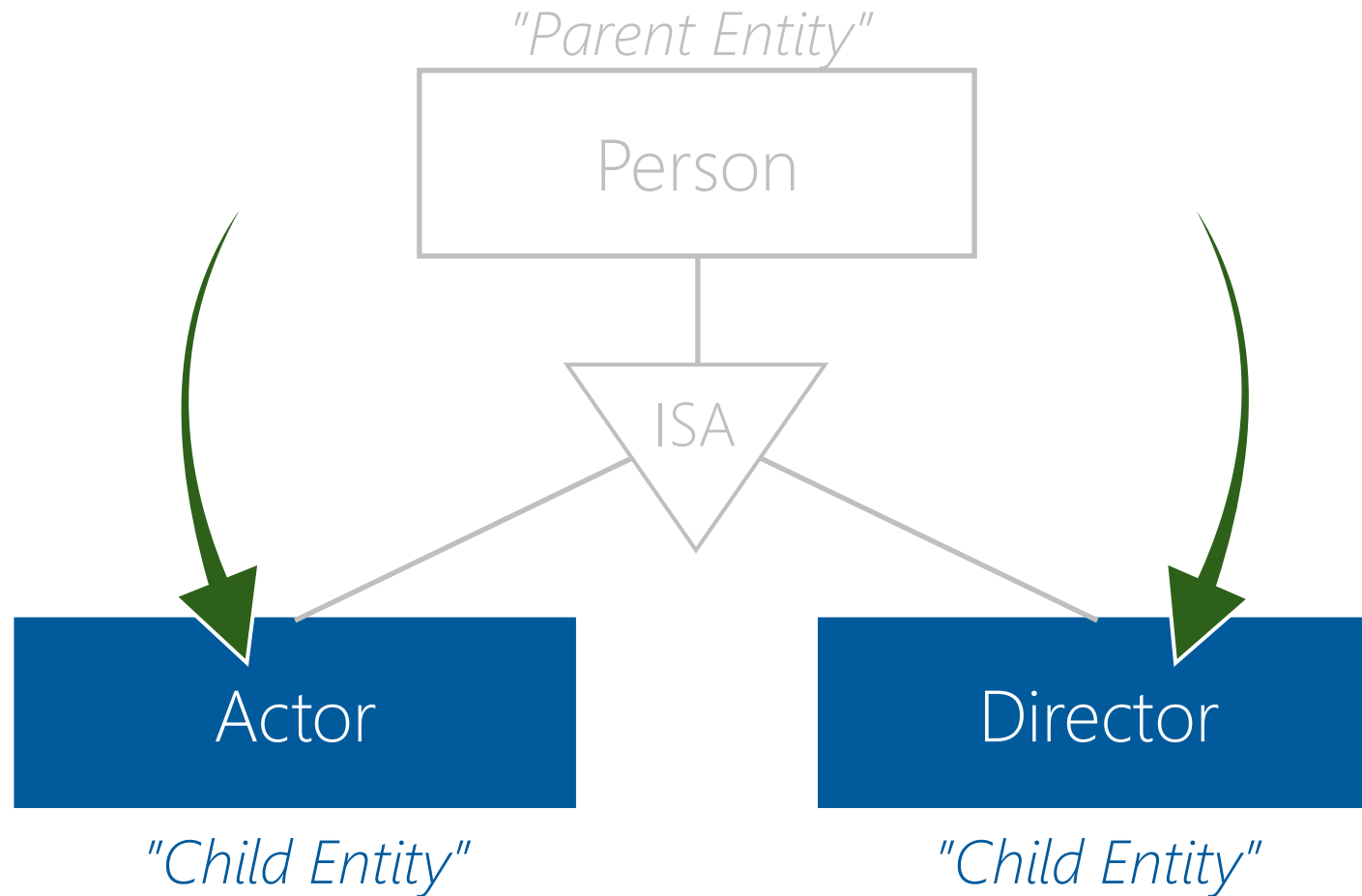
Id	FirstName	LastName	BirthDate	BirthPlace	PersonType	ActorBestLine	ActorMovieCount	DirectorBestMovie	DirectorMovieCount
1	Stanley	Kubrick	Jul. 26, 1928	USA	Director	NULL	NULL	102	13
2	Alfred	Hitchcock	Aug. 13, 1899	England	Director	NULL	NULL	203	47
3	John	Travolta	Feb. 18, 1954	USA	Actor	You ...	61	NULL	NULL
4	Samuel	Jackson	Dec. 21, 1948	USA	Actor	Say 'w...	125	NULL	NULL
5	Uma	Thurman	Apr. 29, 1970	USA	Actor	I believe ..	51	NULL	NULL
6	Hossein	Fani	Mar. 11, 1844	Iran	NULL	NULL	NULL	NULL	NULL
7	Clint	Eastwood	May 31, 1930	USA	ActorDirector	A good ..	69	803	35

- × One big merged relation to have all types of entities.
- × All tuples (entities of different types) share same domains. Actor can have DirectorBestMovie
- × Person, Actor, and Director share same relationship sets.

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IS-A to Relation (Approach II)

Parent spreads into child as if no ISA



IS-A to Relation (Approach II)

Parent spreads into child as if no ISA ... Can it be total and disjoint? Yes.

Id	FirstName	LastName	BirthDate	BirthPlace	DirectorBestMovie	DirectorMovieCount
1	Stanley	Kubrick	Jul. 26, 1928	USA	102	13
2	Alfred	Hitchcock	Aug. 13, 1899	England	203	47

Id	FirstName	LastName	BirthDate	BirthPlace	ActorBestLine	ActorMovieCount
1	John	Travolta	Feb. 18, 1954	USA	You ...	61
2	Samuel	Jackson	Dec. 21, 1948	USA	Say 'w...	125
3	Uma	Thurman	Apr. 29, 1970	USA	I believe ..	51

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IS-A to Relation (Approach II)

Parent spreads into child as if no ISA ... Can it be total and overlapping? Yes.

Id	FirstName	LastName	BirthDate	BirthPlace	DirectorBestMovie	DirectorMovieCount
1	Stanley	Kubrick	Jul. 26, 1928	USA	102	13
2	Alfred	Hitchcock	Aug. 13, 1899	England	203	47
3	Clint	Eastwood	May 31, 1930	USA	803	35

Id	FirstName	LastName	BirthDate	BirthPlace	ActorBestLine	ActorMovieCount
1	John	Travolta	Feb. 18, 1954	USA	You ...	61
2	Samuel	Jackson	Dec. 21, 1948	USA	Say 'w...	125
3	Uma	Thurman	Apr. 29, 1970	USA	I believe ..	51
4	Clint	Eastwood	May 31, 1930	USA	A good ..	69

Redundancy → Update Cascade Anomaly → Inconsistency

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IS-A to Relation (Approach II)

Parent spreads into child as if no ISA ... Can it be partial? Yes, **but new relation.**

Id	FirstName	LastName	BirthDate	BirthPlace	DirectorBestMovie	DirectorMovieCount
1	Stanley	Kubrick	Jul. 26, 1928	USA	102	13
2	Alfred	Hitchcock	Aug. 13, 1899	England	203	47
3	Clint	Eastwood	May 31, 1930	USA	803	35

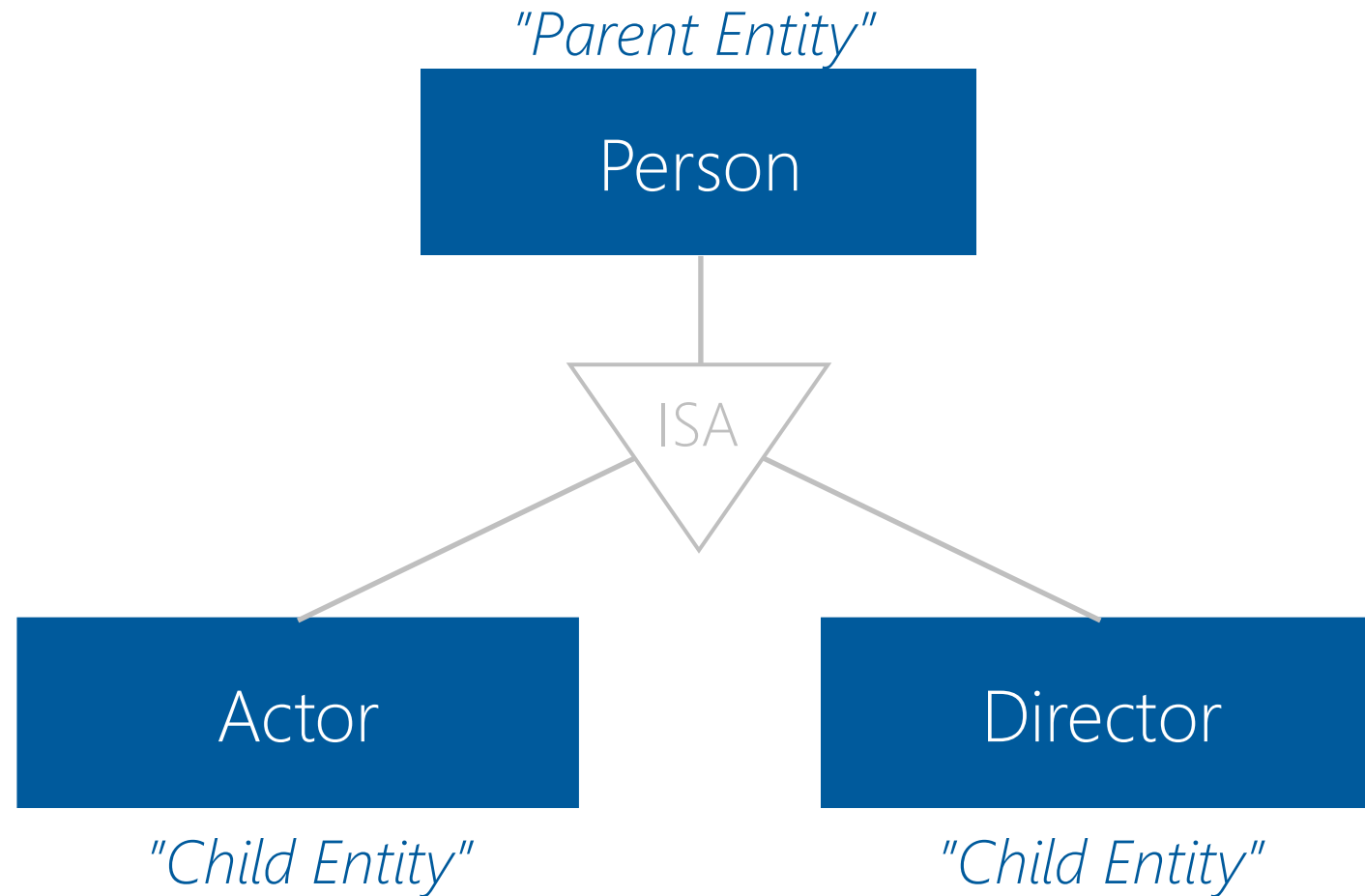
Id	FirstName	LastName	BirthDate	BirthPlace	ActorBestLine	ActorMovieCount
1	John	Travolta	Feb. 18, 1954	USA	You ...	61
2	Samuel	Jackson	Dec. 21, 1948	USA	Say 'w...	125
3	Uma	Thurman	Apr. 29, 1970	USA	I believe ..	51
4	Clint	Eastwood	May 31, 1930	USA	A good ..	69

Id	FirstName	LastName	BirthDate	BirthPlace
1	Hossein	Fani	Mar. 11, 1844	Iran

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IS-A to Relation (Approach III)

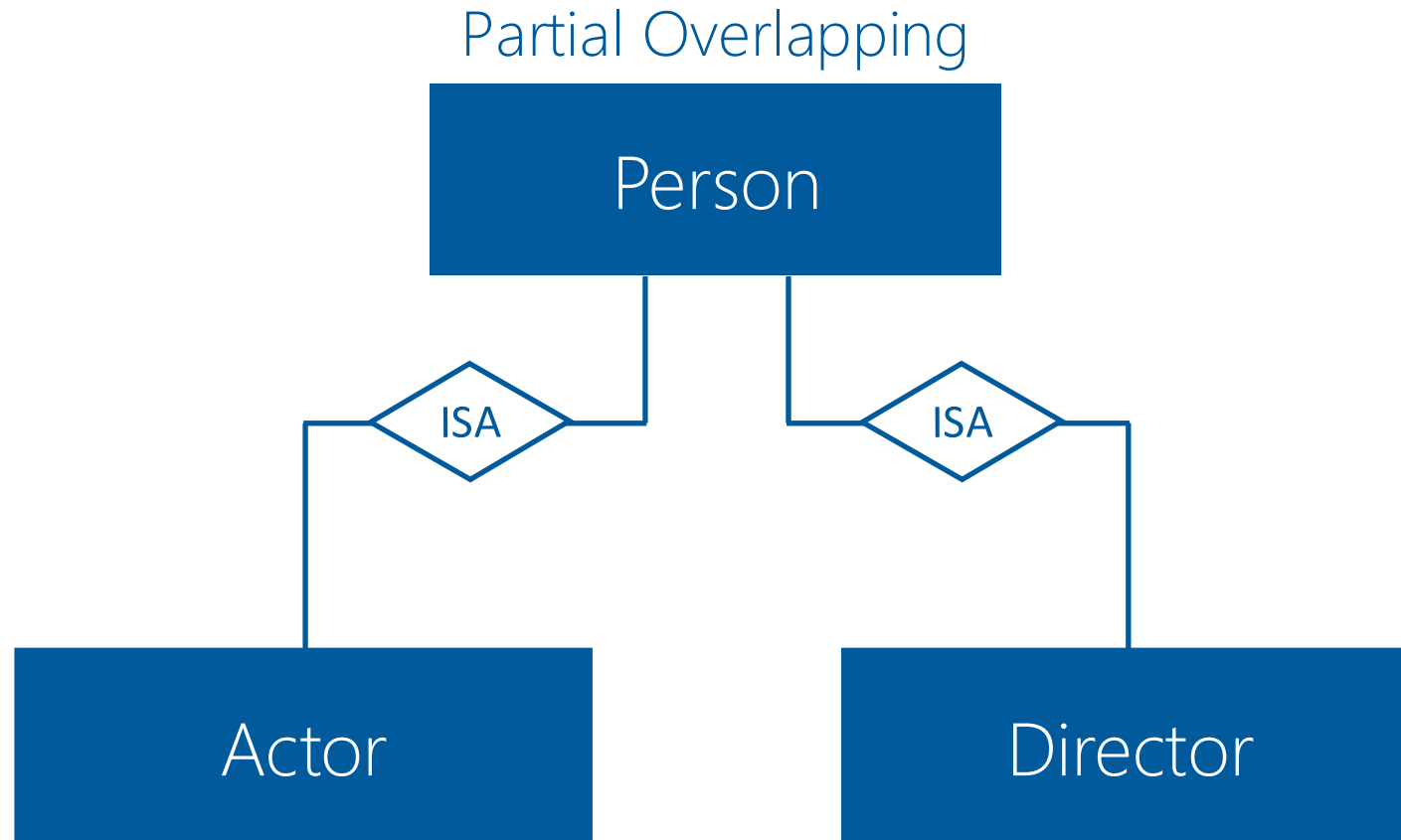
Has its own relation, but connect them using **relationship sets** ...



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IS-A to Relation (Approach III)

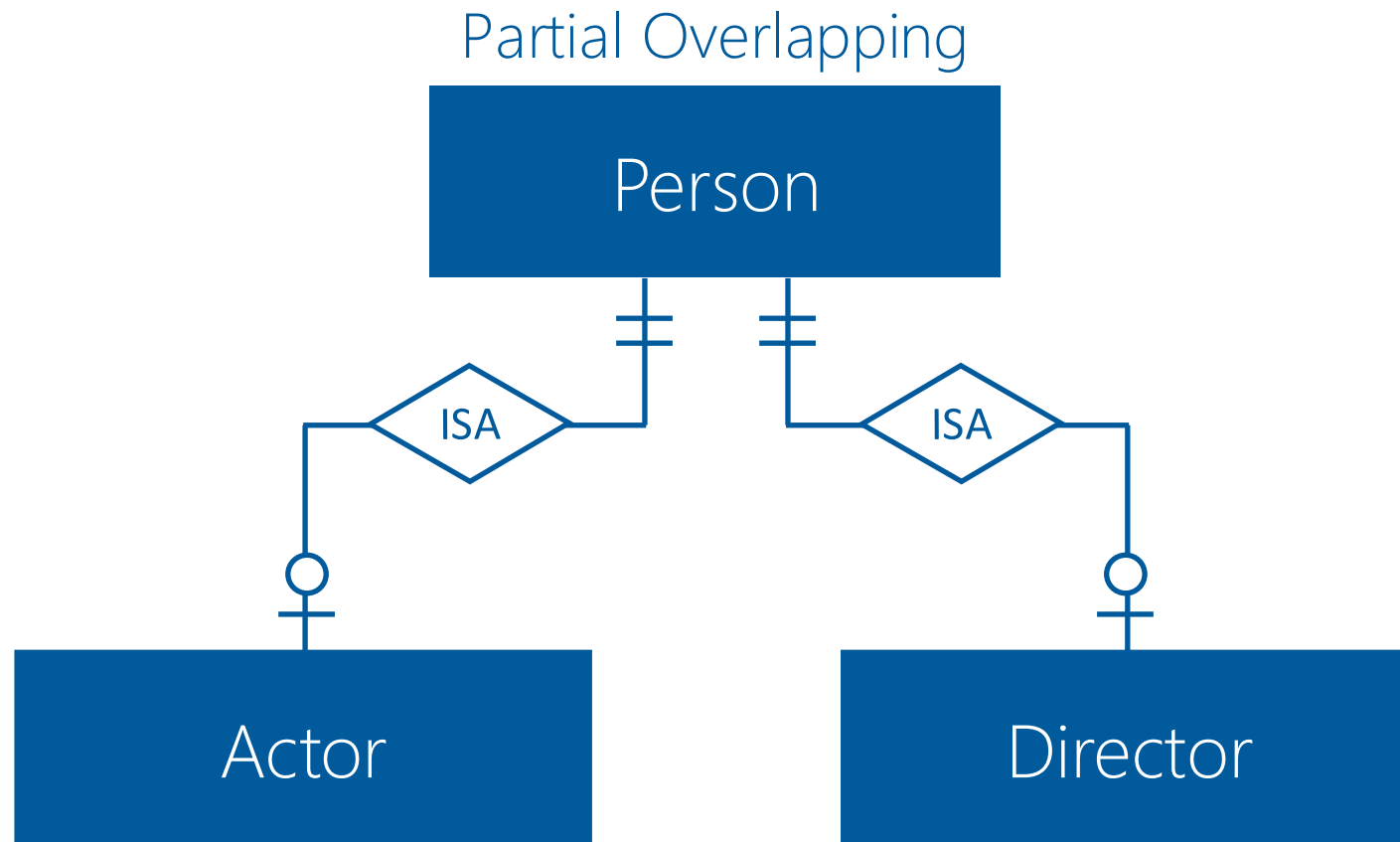
Has its own relation, but connect them using relationship sets



44

IS-A to Relation (Approach III)

Has its own relation, but connect them using relationship sets



45

IS-A to Relation (Approach III)

Has its own relation, but connect them using **one-one** relationship sets

Id	FirstName	LastName	BirthDate	BirthPlace
1	Stanley	Kubrick	Jul. 26, 1928	USA
2	Alfred	Hitchcock	Aug. 13, 1899	England
3	John	Travolta	Feb. 18, 1954	USA
4	Samuel	Jackson	Dec. 21, 1948	USA
5	Uma	Thurman	Apr. 29, 1970	USA
6	Hossein	Fani	Mar. 11, 1844	Iran
7	Clint	Eastwood	May 31, 1930	USA

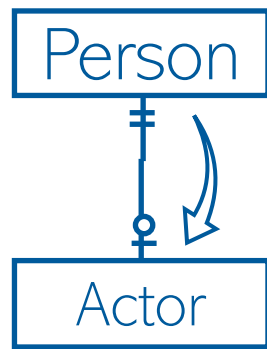
Id	ActorBestLine	ActorMovieCount
1	You ...	61
2	Say 'w...	125
3	I believe ..	51
4	A good ..	69

Id	DirectorBestMovie	DirectorMovieCount
1	102	13
2	203	47
3	803	35

46

IS-A to Relation (Approach III-A)

Has its own relation, but connect them using one-one relationship sets



Id	FirstName	LastName	BirthDate	BirthPlace
1	Stanley	Kubrick	Jul. 26, 1928	USA
2	Alfred	Hitchcock	Aug. 13, 1899	England
3	John	Travolta	Feb. 18, 1954	USA
4	Samuel	Jackson	Dec. 21, 1948	USA
5	Uma	Thurman	Apr. 29, 1970	USA
6	Hossein	Fani	Mar. 11, 1844	Iran
7	Clint	Eastwood	May 31, 1930	USA

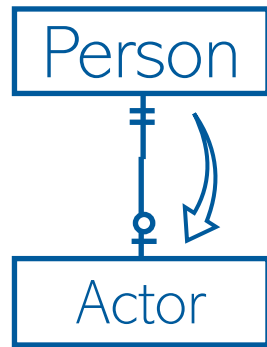
FK=CK=Person.Id, Mandatory

Id	ActorBestLine	ActorMovieCount	PersonId
1	You ...	61	3
2	Say 'w...	125	4
3	I believe ..	51	5
4	A good ..	69	7

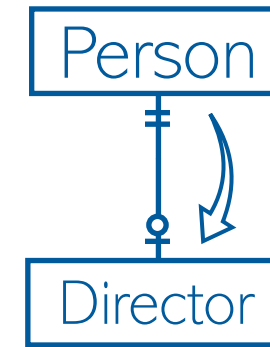
47

IS-A to Relation (Approach III-A)

Has its own relation, but connect them using one-one relationship sets



Id	FirstName	LastName	BirthDate	BirthPlace
1	Stanley	Kubrick	Jul. 26, 1928	USA
2	Alfred	Hitchcock	Aug. 13, 1899	England
3	John	Travolta	Feb. 18, 1954	USA
4	Samuel	Jackson	Dec. 21, 1948	USA
5	Uma	Thurman	Apr. 29, 1970	USA
6	Hossein	Fani	Mar. 11, 1844	Iran
7	Clint	Eastwood	May 31, 1930	USA



FK=CK=Person.Id, Mandatory

Id	ActorBestLine	ActorMovieCount	PersonId
1	You ...	61	3
2	Say 'w...	125	4
3	I believe ..	51	5
4	A good ..	69	7

FK=CK=Person.Id, Mandatory

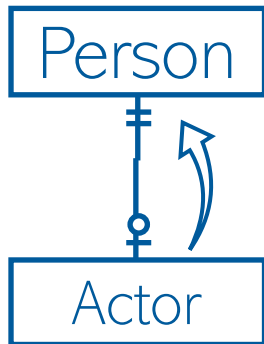
Id	DirectorBestMovie	DirectorMovieCount	PersonId
1	102	13	1
2	203	47	2
3	803	35	7

48

IS-A to Relation (Approach III-B)

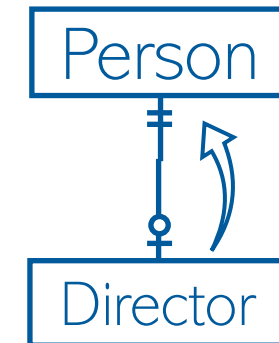
Has its own relation, but connect them using one-one relationship sets

Id	FirstName	LastName	BirthDate	BirthPlace
1	Stanley	Kubrick	Jul. 26, 1928	USA
2	Alfred	Hitchcock	Aug. 13, 1899	England
3	John	Travolta	Feb. 18, 1954	USA
4	Samuel	Jackson	Dec. 21, 1948	USA
5	Uma	Thurman	Apr. 29, 1970	USA
6	Hossein	Fani	Mar. 11, 1844	Iran
7	Clint	Eastwood	May 31, 1930	USA



Id	ActorBestLine	ActorMovieCount
1	You ...	61
2	Say 'w...	125
3	I believe ..	51
4	A good ..	69

Id	DirectorBestMovie	DirectorMovieCount
1	102	13
2	203	47
3	803	35



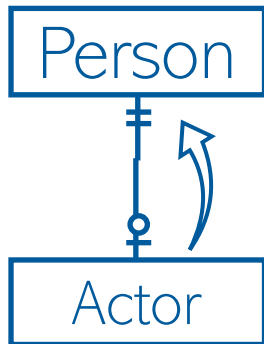
49

IS-A to Relation (Approach III-B)

Has its own relation, but connect them using one-one relationship sets

Id	FirstName	LastName	BirthDate	BirthPlace	ActorId
1	Stanley	Kubrick	Jul. 26, 1928	USA	NULL
2	Alfred	Hitchcock	Aug. 13, 1899	England	NULL
3	John	Travolta	Feb. 18, 1954	USA	1
4	Samuel	Jackson	Dec. 21, 1948	USA	2
5	Uma	Thurman	Apr. 29, 1970	USA	3
6	Hossein	Fani	Mar. 11, 1844	Iran	NULL
7	Clint	Eastwood	May 31, 1930	USA	4

$FK_1 = CK_1 = \text{Actor.Id}$, Optional



Id	ActorBestLine	ActorMovieCount
1	You ...	61
2	Say 'w...	125
3	I believe ..	51
4	A good ..	69

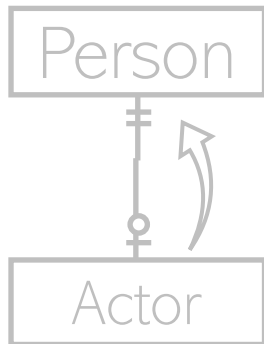
50

IS-A to Relation (Approach III-B)

Has its own relation, but connect them using one-one relationship sets

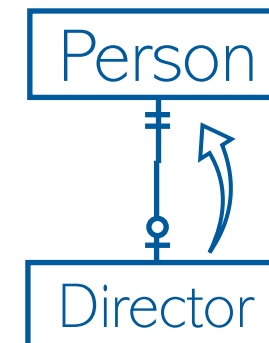
Id	FirstName	LastName	BirthDate	BirthPlace	ActorId	DirectorId
1	Stanley	Kubrick	Jul. 26, 1928	USA	NULL	1
2	Alfred	Hitchcock	Aug. 13, 1899	England	NULL	2
3	John	Travolta	Feb. 18, 1954	USA	1	NULL
4	Samuel	Jackson	Dec. 21, 1948	USA	2	NULL
5	Uma	Thurman	Apr. 29, 1970	USA	3	NULL
6	Hossein	Fani	Mar. 11, 1844	Iran	NULL	NULL
7	Clint	Eastwood	May 31, 1930	USA	4	3

$FK_1 = CK_1 = \text{Actor.Id}$, Optional
 $FK_2 = CK_2 = \text{Director.Id}$, Optional



Id	ActorBestLine	ActorMovieCount
1	You ...	61
2	Say 'w...	125
3	I believe ..	51
4	A good ..	69

Id	DirectorBestMovie	DirectorMovieCount
1	102	13
2	203	47
3	803	35



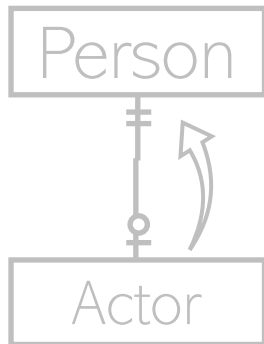
51

IS-A to Relation (Approach III-B)

Has its own relation, but connect them using one-one relationship sets

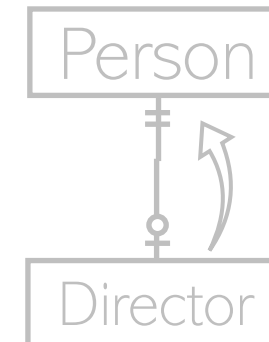
Id	FirstName	LastName	BirthDate	BirthPlace	ActorId	DirectorId
1	Stanley	Kubrick	Jul. 26, 1928	USA	NULL	1
2	Alfred	Hitchcock	Aug. 13, 1899	England	NULL	2
3	John	Travolta	Feb. 18, 1954	USA	1	NULL
4	Samuel	Jackson	Dec. 21, 1948	USA	2	NULL
5	Uma	Thurman	Apr. 29, 1970	USA	3	NULL
6	Hossein	Fani	Mar. 11, 1844	Iran	NULL	NULL
7	Clint	Eastwood	May 31, 1930	USA	4	3

$FK_1 = CK_1 = \text{Actor.Id}$, Optional
 $FK_2 = CK_2 = \text{Director.Id}$, Optional



Id	ActorBestLine	ActorMovieCount
1	You ...	61
2	Say 'w...	125
3	I believe ..	51
4	A good ..	69

Id	DirectorBestMovie	DirectorMovieCount
1	102	13
2	203	47
3	803	35



52

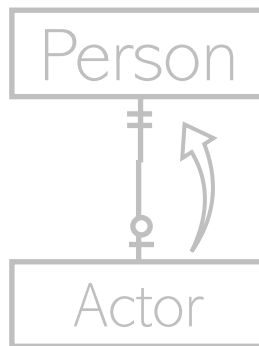
IS-A to Relation (Approach III-B)

Has its own relation, but connect them using one-one relationship sets

Partial & Overlapping.
Partial & Disjoint?
Total & Overlapping?
Total & Disjoint?

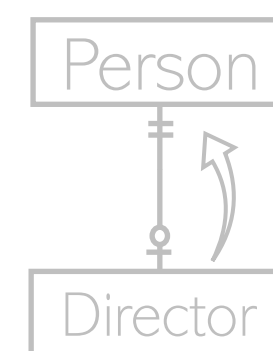
Id	FirstName	LastName	BirthDate	BirthPlace	ActorId	DirectorId
1	Stanley	Kubrick	Jul. 26, 1928	USA	NULL	1
2	Alfred	Hitchcock	Aug. 13, 1899	England	NULL	2
3	John	Travolta	Feb. 18, 1954	USA	1	NULL
4	Samuel	Jackson	Dec. 21, 1948	USA	2	NULL
5	Uma	Thurman	Apr. 29, 1970	USA	3	NULL
6	Hossein	Fani	Mar. 11, 1844	Iran	NULL	NULL
7	Clint	Eastwood	May 31, 1930	USA	4	3

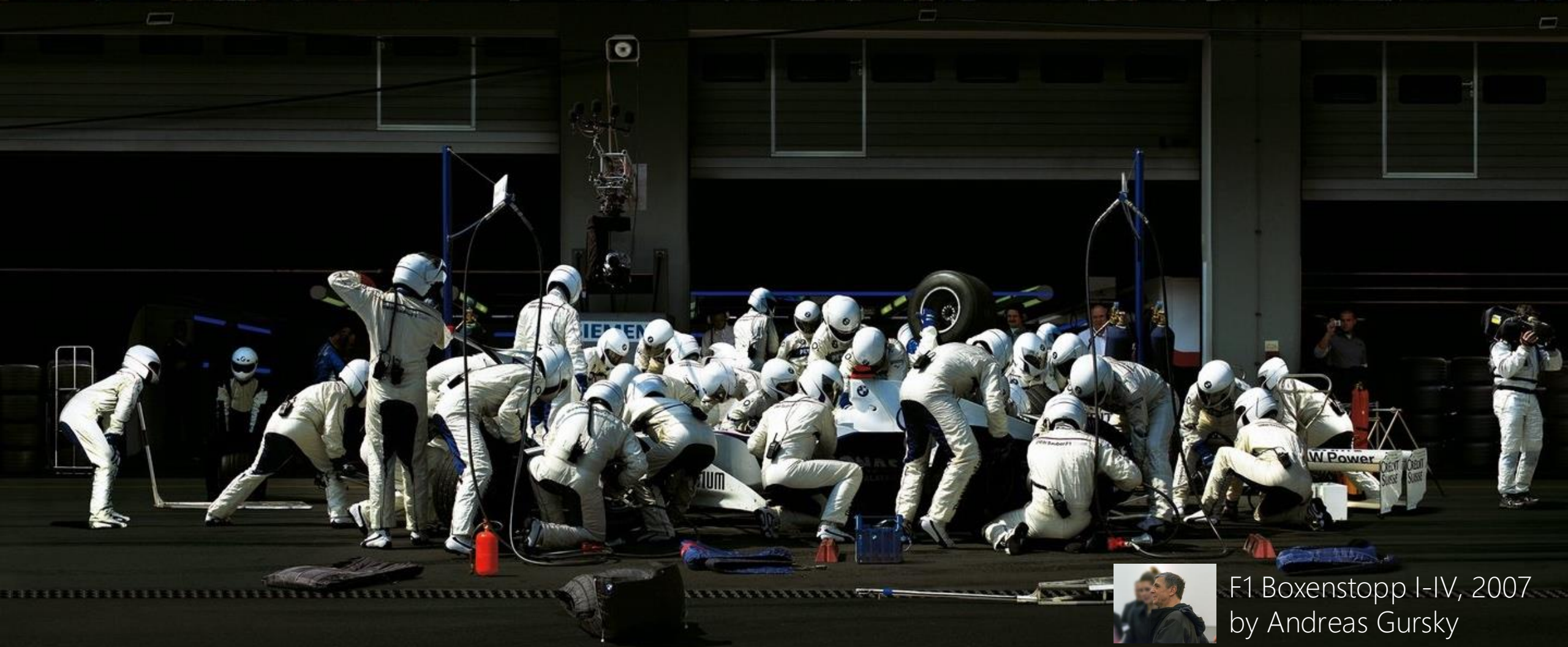
$FK_1 = CK_1 = \text{Actor.Id}$, Optional
 $FK_2 = CK_2 = \text{Director.Id}$, Optional



Id	ActorBestLine	ActorMovieCount
1	You ...	61
2	Say 'w...	125
3	I believe ..	51
4	A good ..	69

Id	DirectorBestMovie	DirectorMovieCount
1	102	13
2	203	47
3	803	35

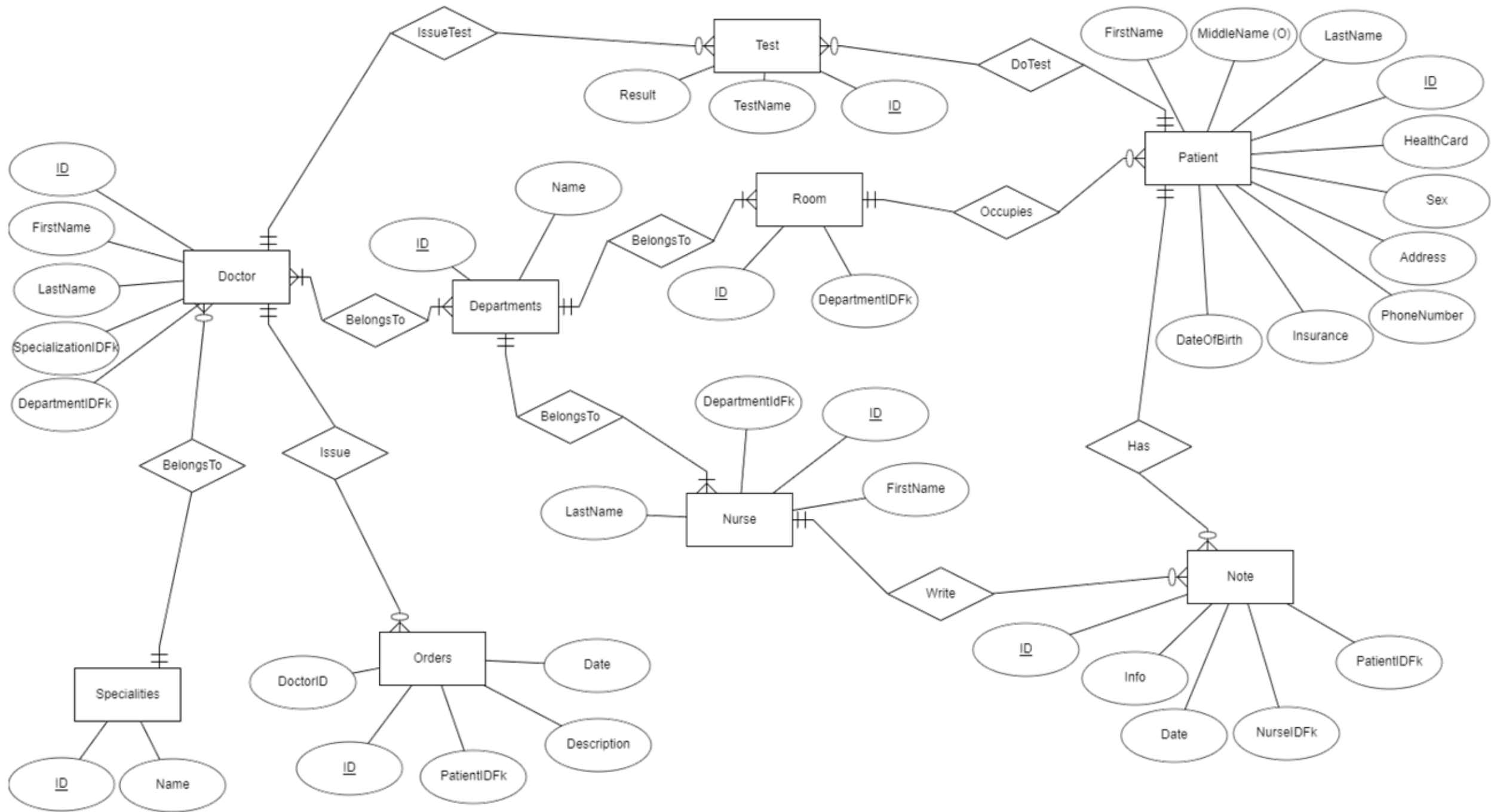




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by Andreas Gursky

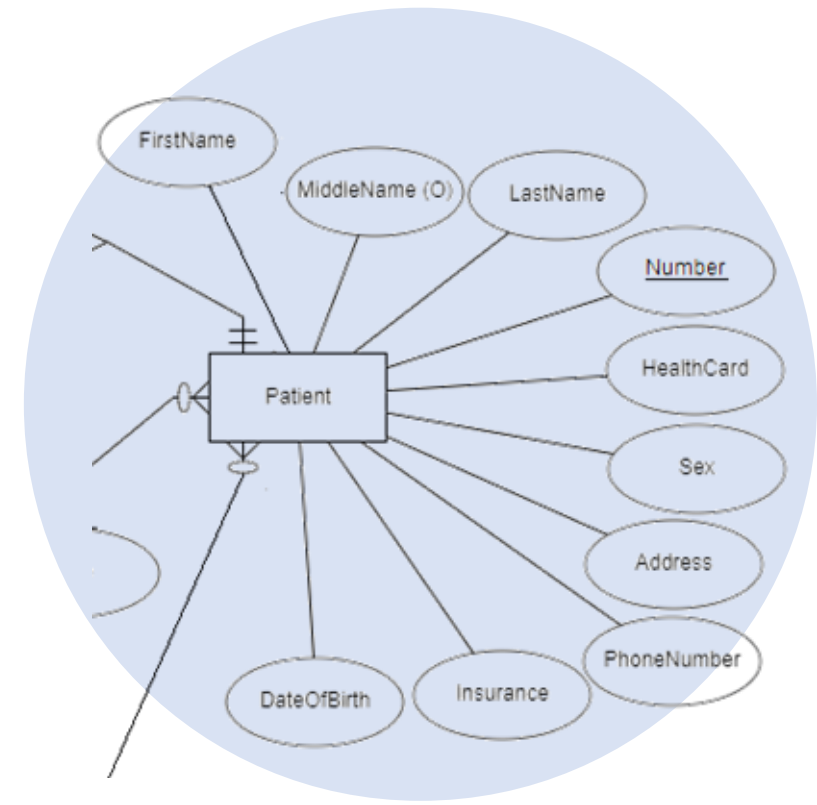


by Andreas Gursky, \$4.3 million, 2nd Most Expensive Photo
https://en.wikipedia.org/wiki/Rhein_II



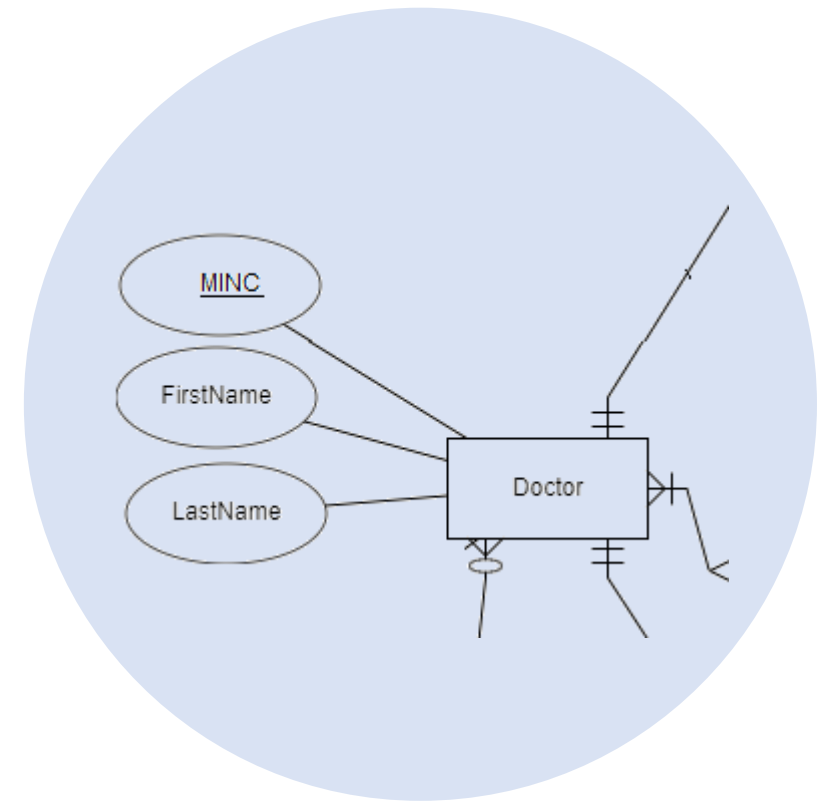
R1: Patient(Id,
Number,
FirstName,
MiddleName,
LastName,
Sex,
DateOfBirth,
Address,
Phone,
HealthCardNo,
InsuranceName)

CK={Number} & Mandatory



R2: Doctor(Id,
MINC,
FirstName,
LastName)

CK={MINC} & Mandatory



R3: Room(Id, Number)
CK={Number} & Mandatory

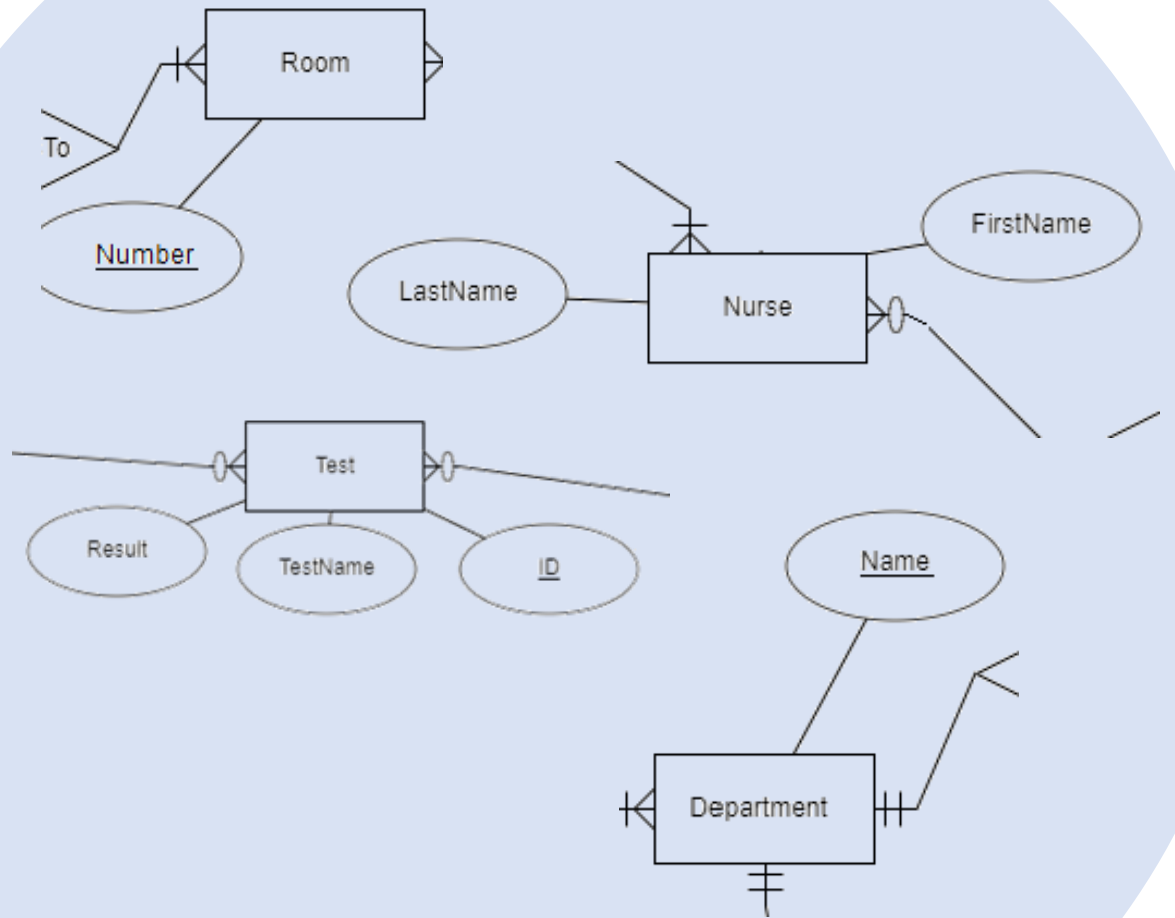
R4: Nurse(Id, FirstName, LastName)

R5: Test(Id, Name, Result)
CK={Name} & Mandatory

R6: Order(Id, Description, Date)

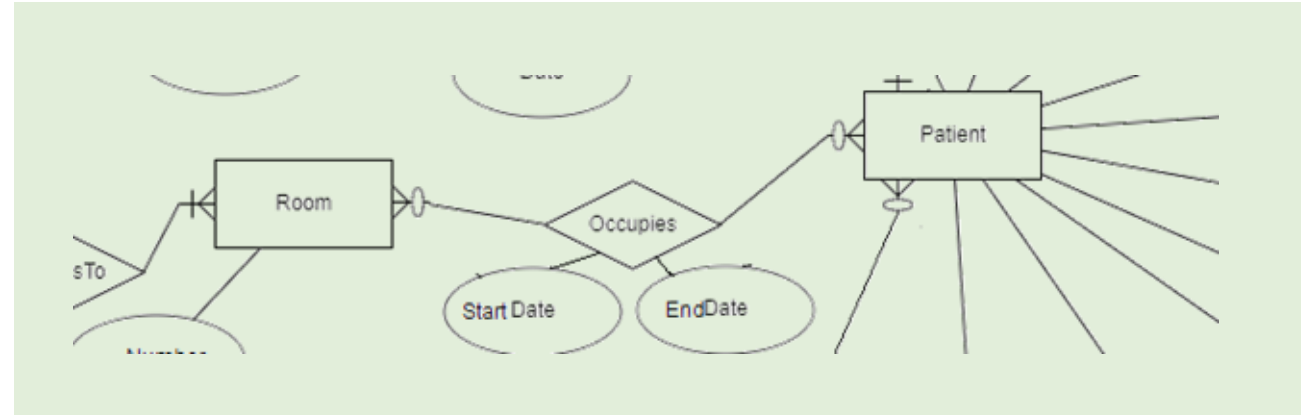
R7: Specialty(Id, Name)
CK={Name} & Mandatory

R8: Department(Id, Name)
CK={Name} & Mandatory

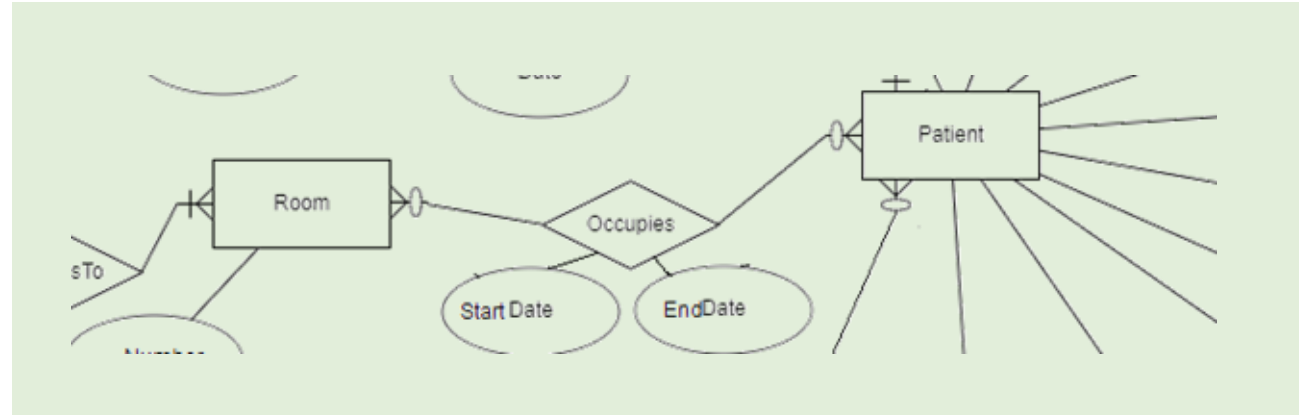


R9: PatientRoom(Id,
PatientId,
RoomId,
StartDate,
EndDate)

$FK_1 = \{PatientId\}$ & Mandatory
 $FK_2 = \{RoomId\}$ & Mandatory
 $CK = \{PatientId, RoomId\}$



R9: PatientRoom(Id,
PatientId,
RoomId,
StartDate,
EndDate)



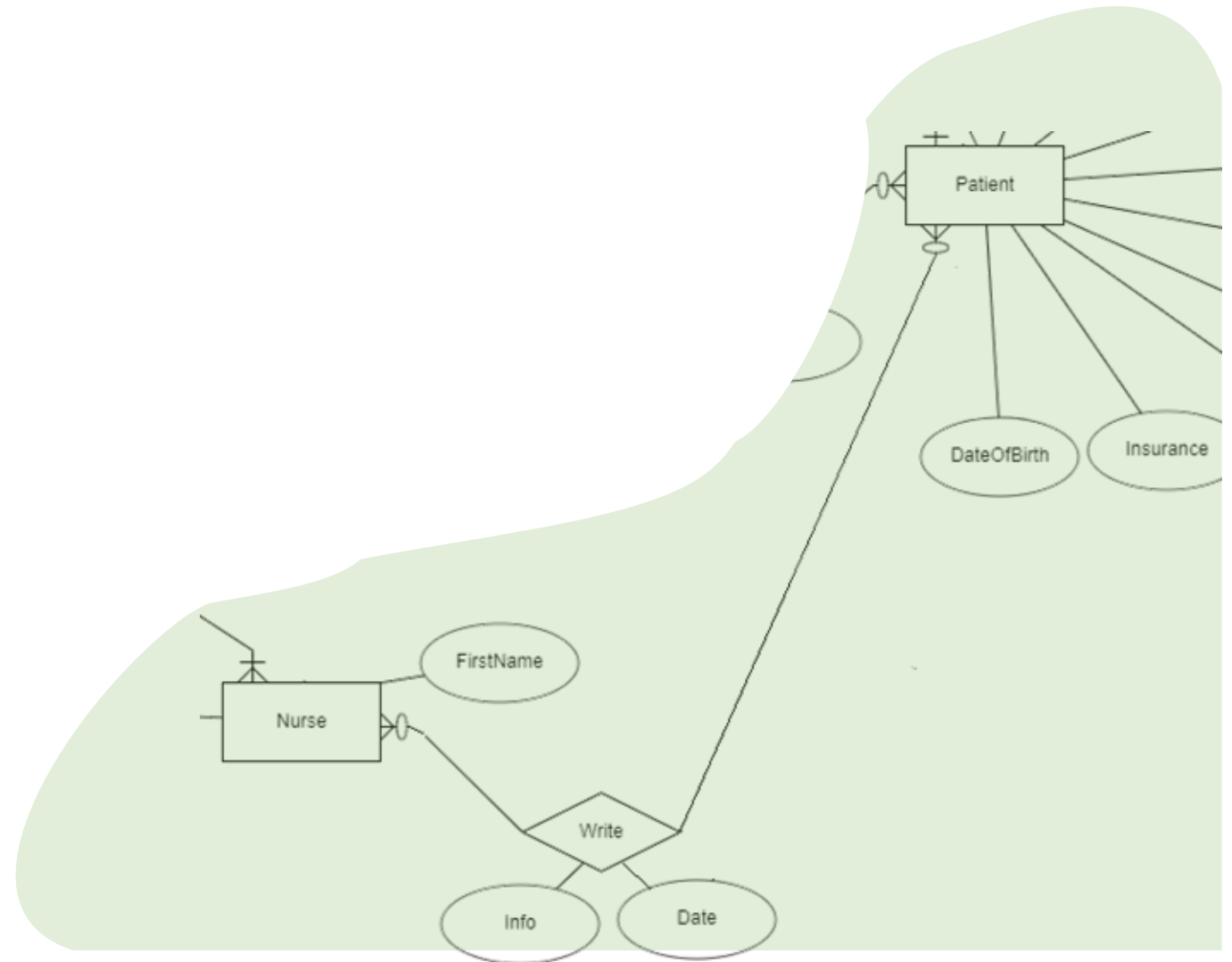
$FK_1 = \{PatientId\}$ & Mandatory

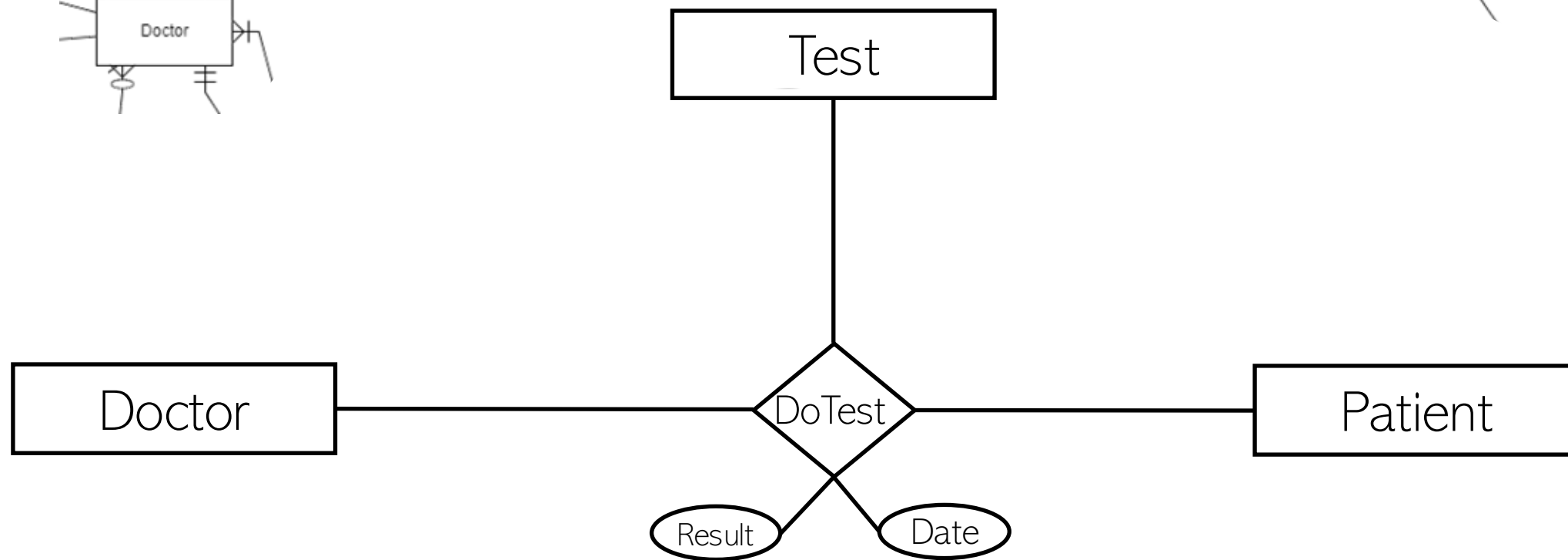
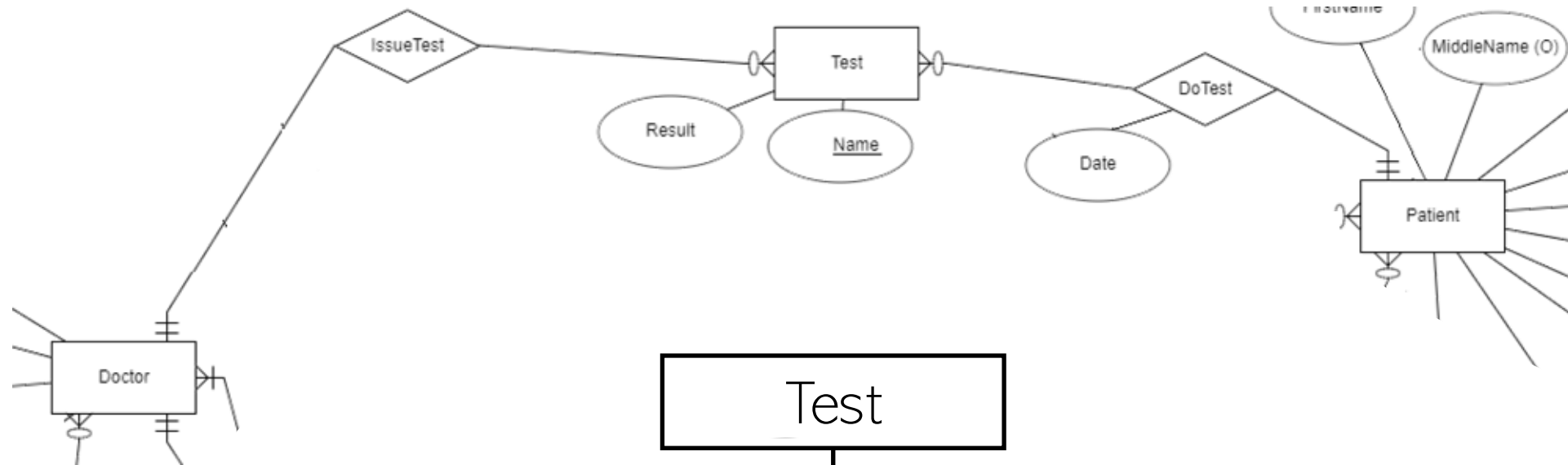
$FK_2 = \{RoomId\}$ & Mandatory

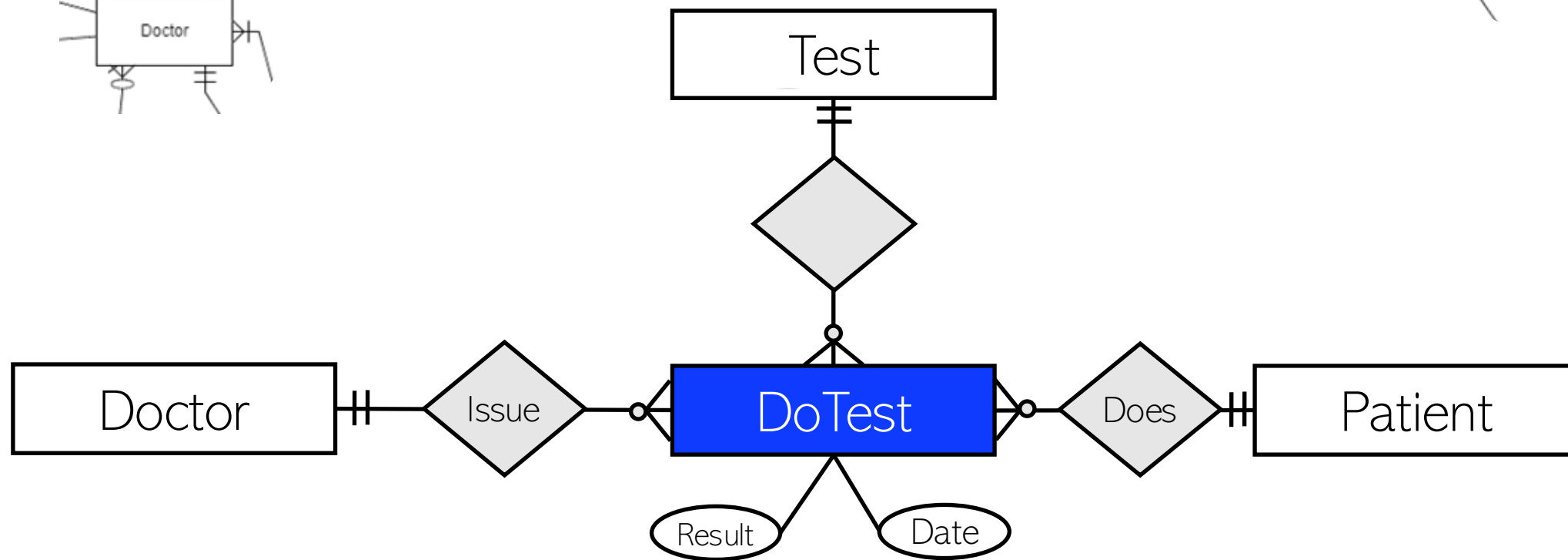
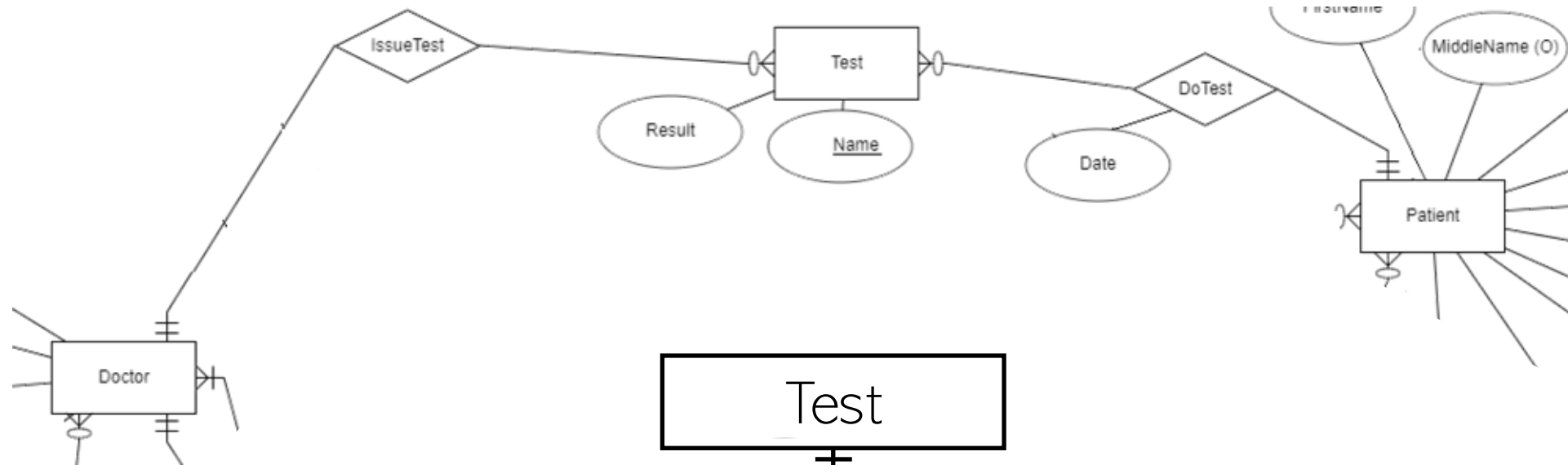
$CK = \{PatientId, RoomId\} \cup \{StartDate\}$ (Why?)

R10: PatientNurse(Id,
PatientId,
Nurseld,
Date,
Info)

$FK_1 = \{PatientId\}$ & Mandatory
 $FK_2 = \{Nurseld\}$ & Mandatory
 $CK = \{PatientId, Nurseld, Date\}$

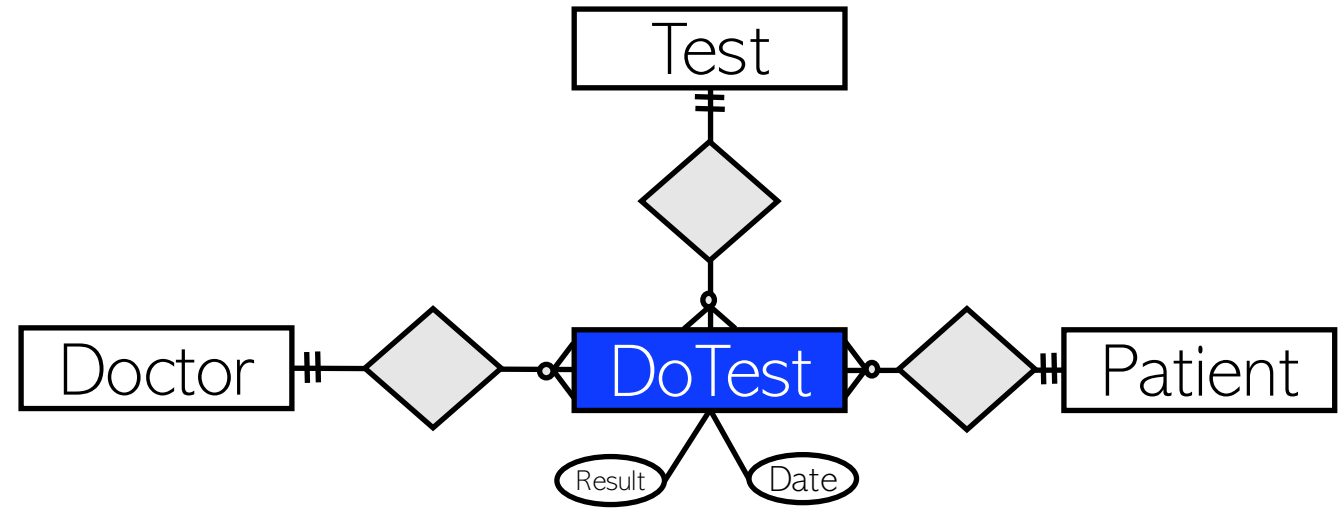






R5: Test(Id, Name)

CK={Name} & Mandatory



R11: DoTest(Id, DoctorId, PatientId, TestId, Result, Date)

FK₁= {PatientId} & Mandatory

FK₂= {DoctorId} & Mandatory

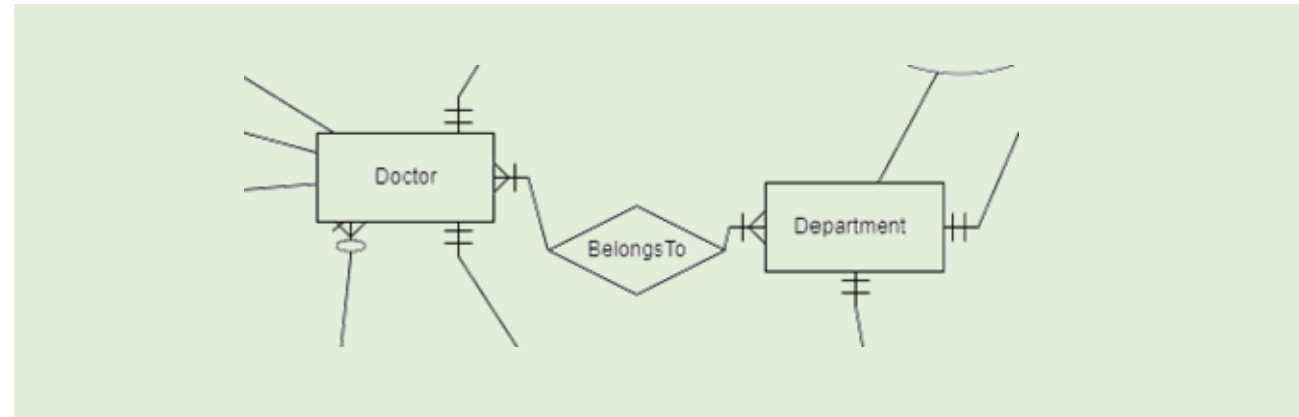
FK₃= {TestId} & Mandatory

R12: DoctorDepartment(Id, DoctorId, DepartmentId)

FK1={DoctorId} & Mandatory

FK2={DepartmentId} & Mandatory

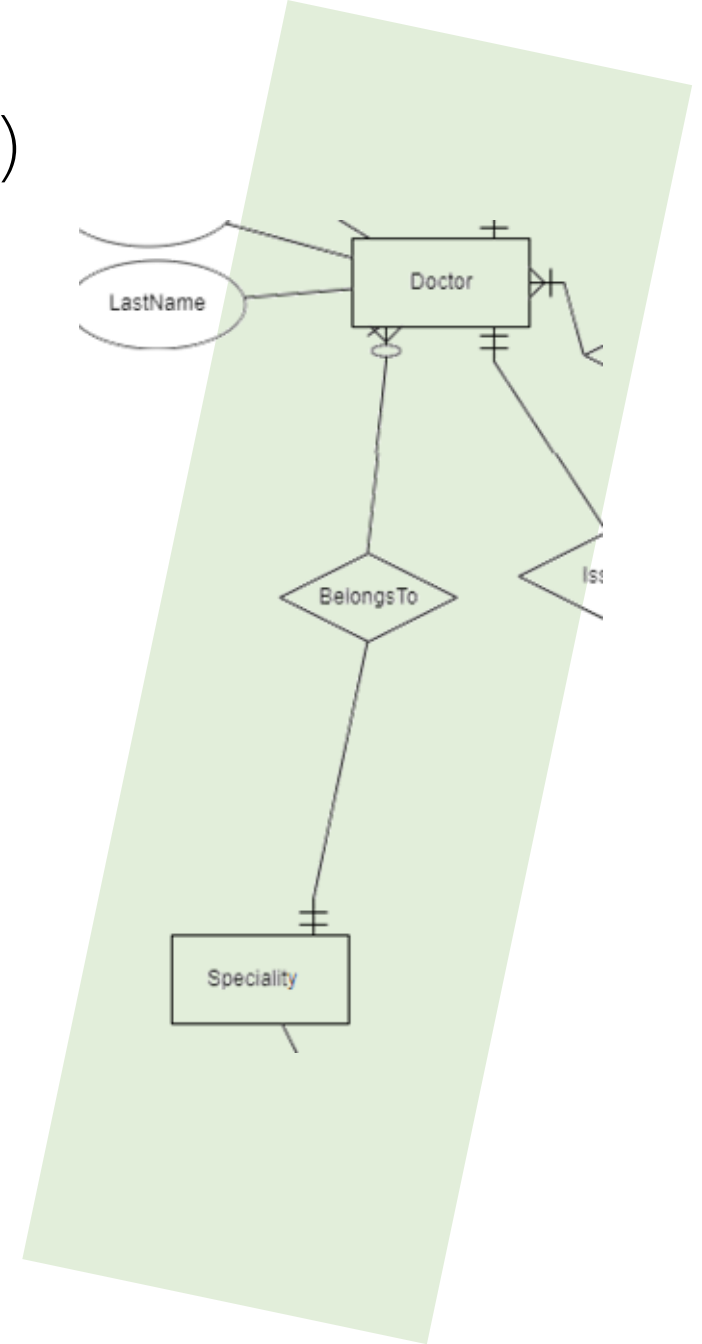
CK={DoctorId, DepartmentId}



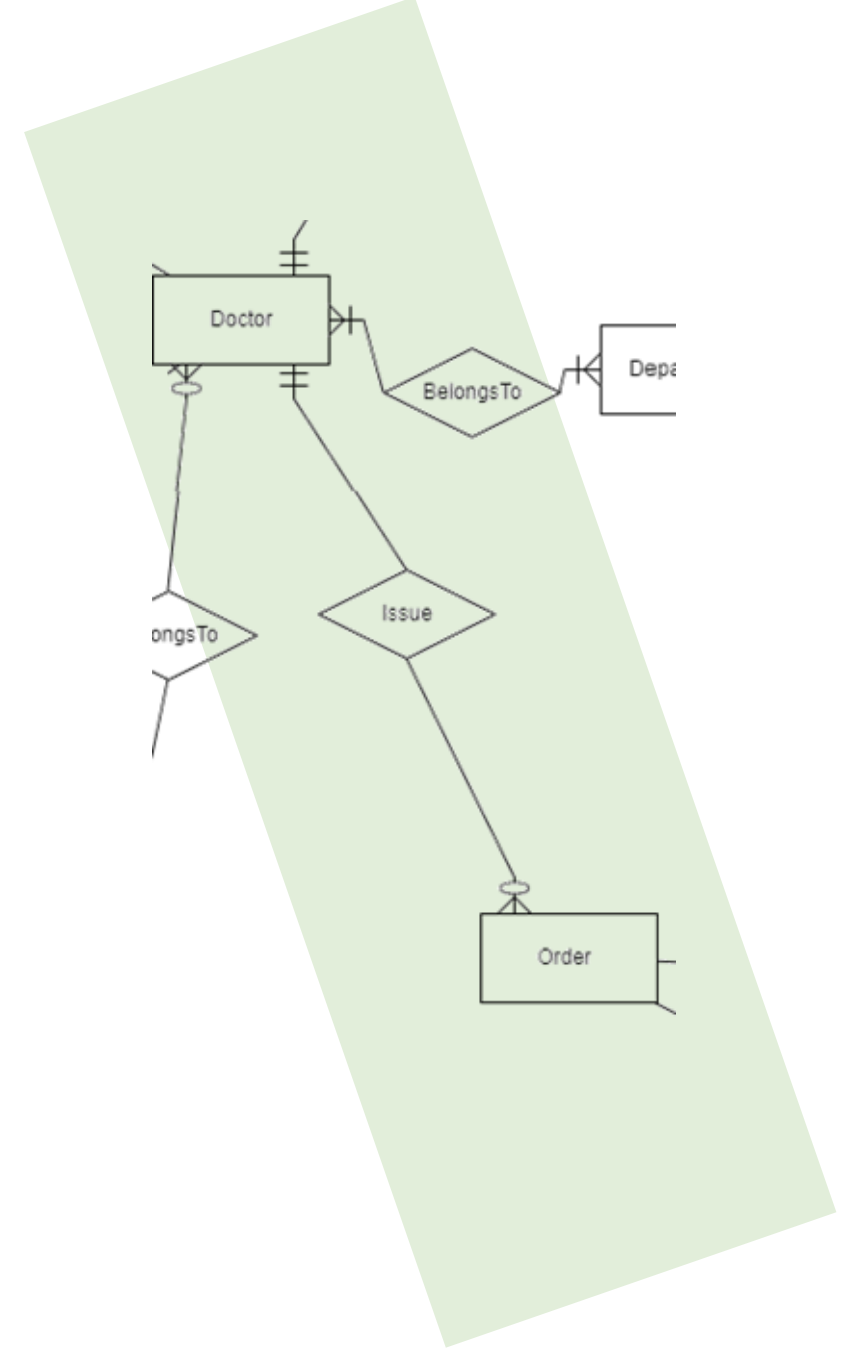
R2: Doctor(Id, MINC, FirstName, LastName, SpecialtyId)

CK={MINC} & Mandatory

FK={SpecialtyId} & Mandatory



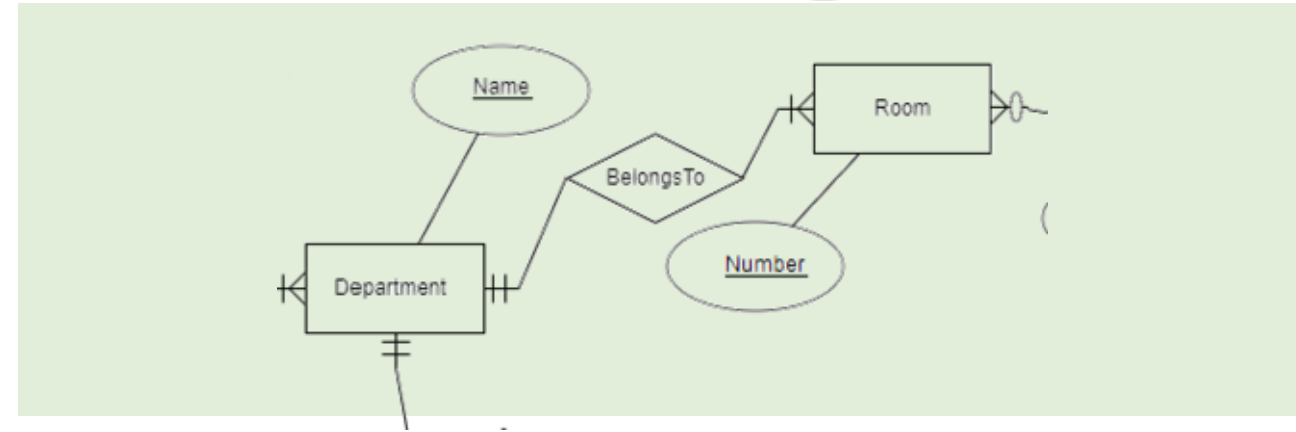
R6: Order(Id, Description, Date, DoctorId)
FK={DoctorId} & Mandatory



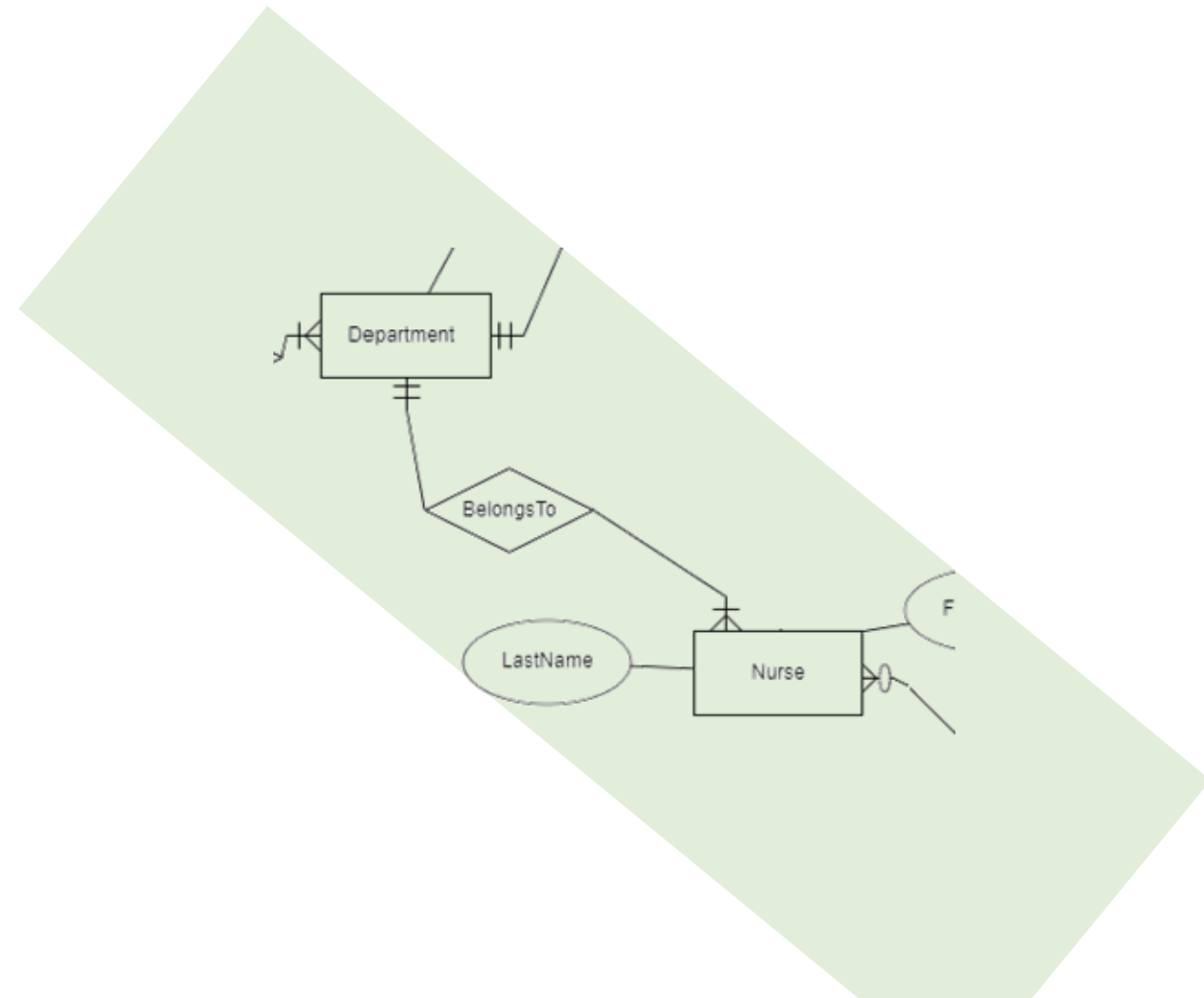
R3: Room(Id, Number, DepartmentId)

CK={Number} & Mandatory

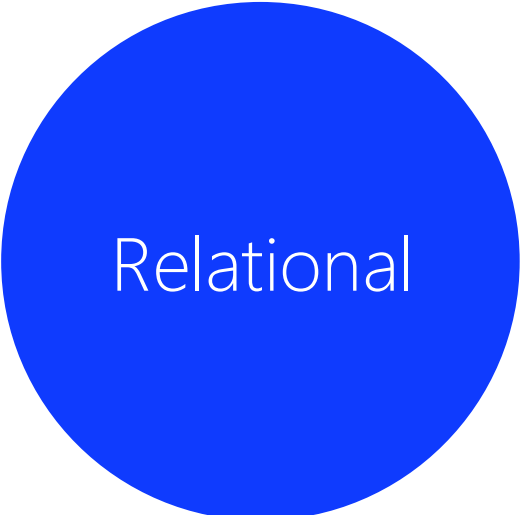
FK={DepartmentId} & Mandatory



R4: Nurse(Id, FirstName, LastName, DepartmentId)
FK={DepartmentId} & Mandatory



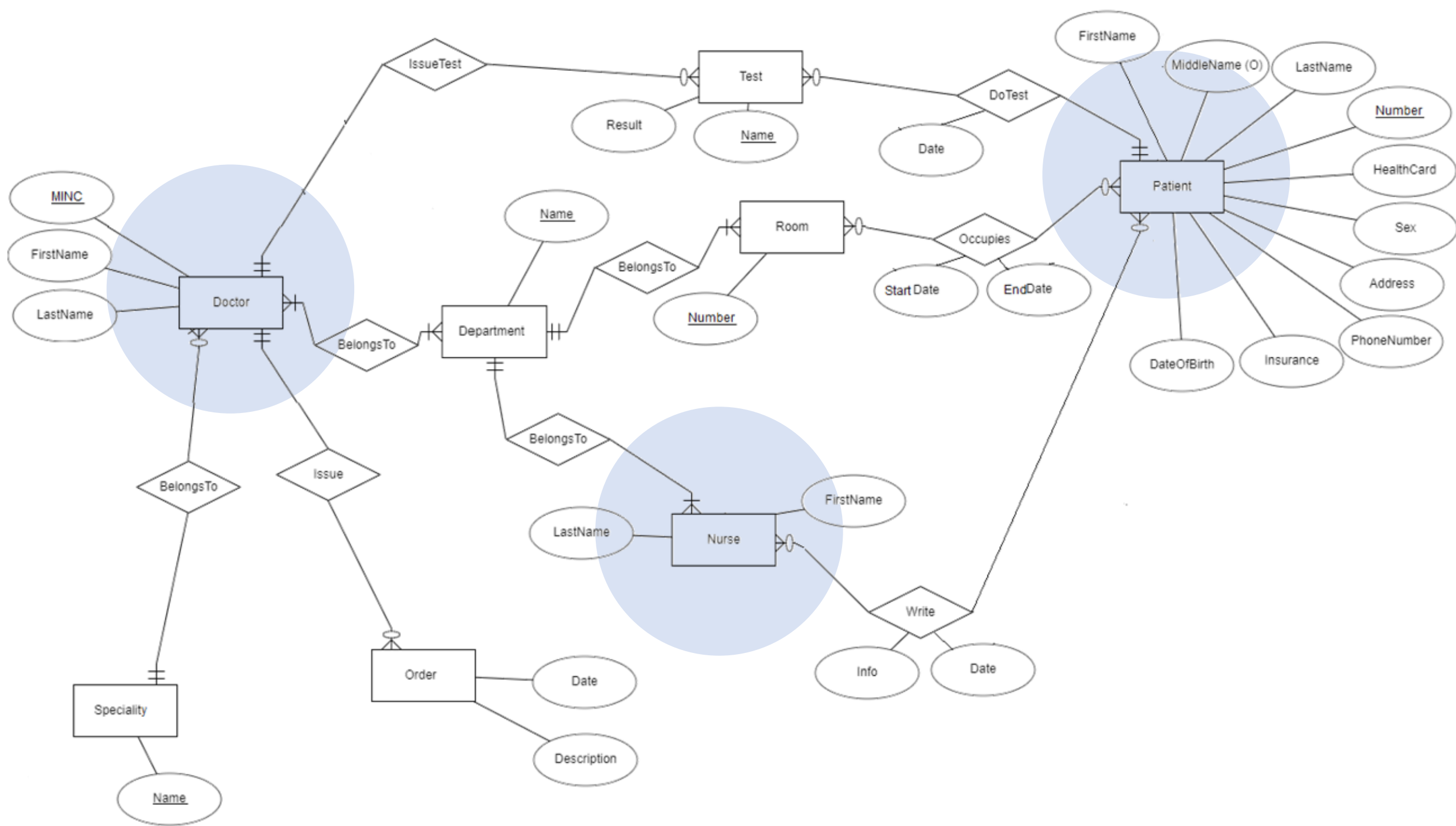
R1: Patient(Id, Number, FirstName, MiddleName, LastName, Sex, DateOfBirth, Address, Phone, HealthCardNo, InsuranceName), CK={Number} & Mandatory
R2: Doctor(Id, (MINC, FirstName, LastName, SpecialtyId), CK={MINC} & Mandatory, FK={SpecialtyId} & Mandatory
R3: Room(Id, Number, DepartmentId), CK={Number} & Mandatory, FK={DepartmentId} & Mandatory
R4: Nurse(Id, FirstName, LastName, DepartmentId), FK={DepartmentId} & Mandatory
R5: Test(Id, Name), CK={Name} & Mandatory
R6: Order(Id, Description, Date, DoctorId), FK={DoctorId} & Mandatory
R7: Specialty(Id, Name), CK={Name} & Mandatory
R8: Department(Id, Name), CK={Name} & Mandatory
R9: PatientRoom(Id, PatientId, RoomId, StartDate, EndDate), FK1={PatientId}, FK2={RoomId}, CK={PatientId, RoomId, StartDate}
R10: PatientNurse(Id, PatientId, Nurseld, Date, Info), FK1={PatientId}, FK2={Nurseld}, CK={PatientId, Nurseld, Date}
R11: DoTest(Id, DoctorId, PatientId, TestId, Result, Date), FK1={PatientId} & Mandatory, FK2={DoctorId} & Mandatory, FK3={PatientId} & Mandatory, CK={DoctorId, PatientId, TestId, Date}
R12: DoctorDepartment(Id, DoctorId, DepartmentId), FK1={DoctorId}, FK2={DepartmentId}, CK={DoctorId, DepartmentId}

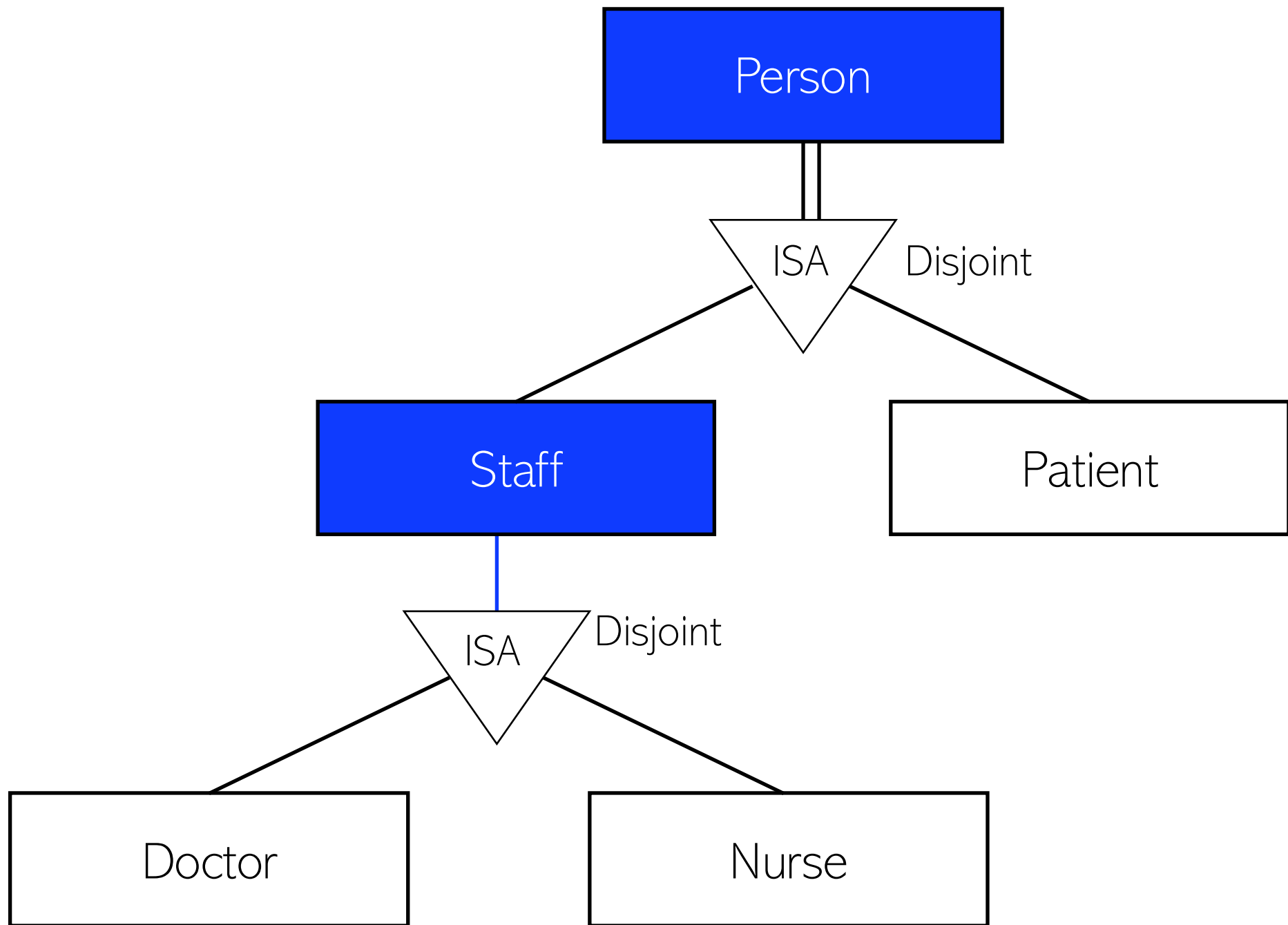


Relational

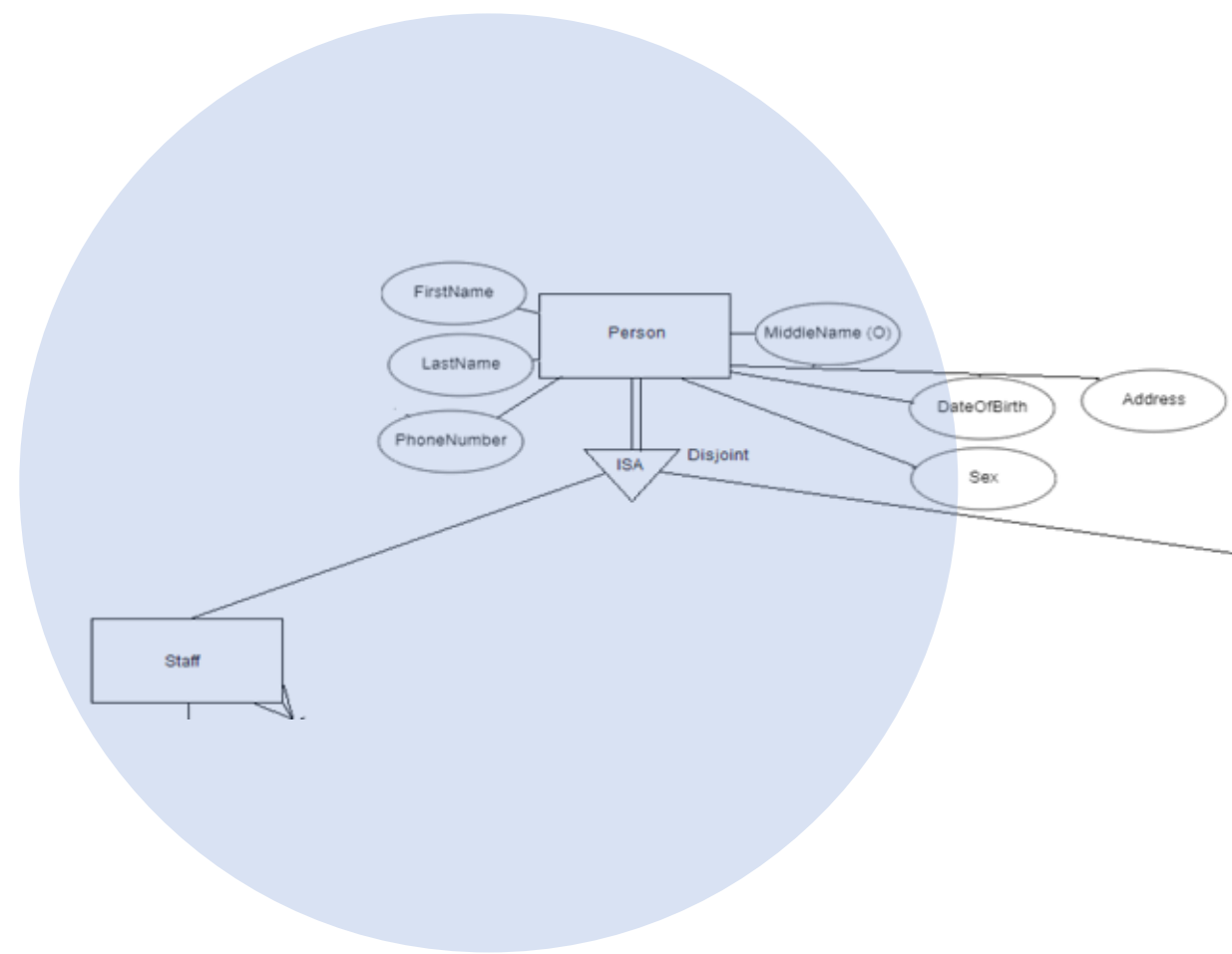


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R13: Person(Id,
 FirstName,
 MiddleName,
 LastName,
 Sex,
 DateOfBirth,
 Address,
 Phone)



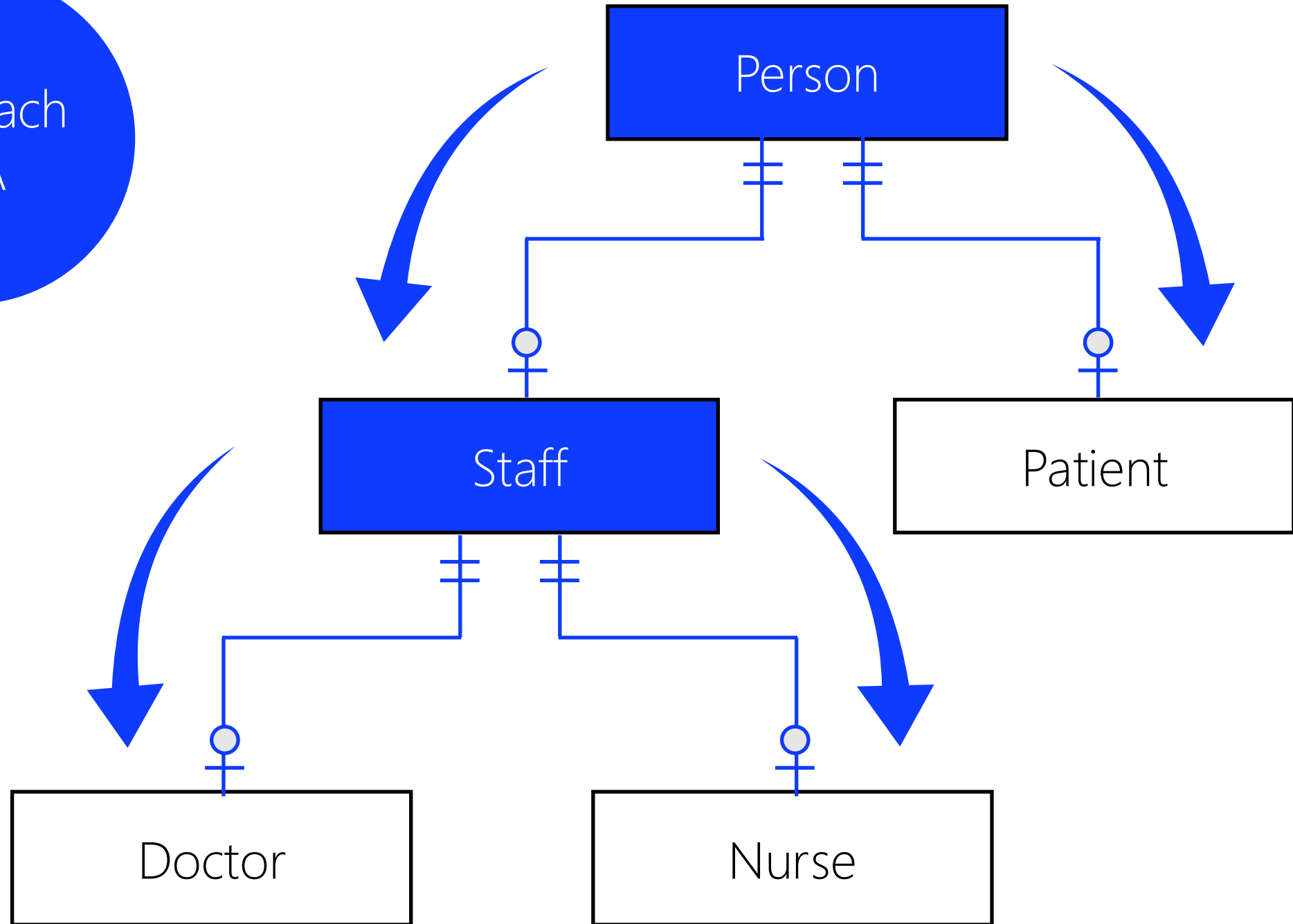
R14: Staff(Id)

R1: Patient(Id, Number, HealthCardNo, InsuranceName), CK={Number} & Mandatory

R2: Doctor(Id, MINC, SpecialtyId), CK={MINC} & Mandatory, FK={SpecialtyId} & Mandatory

R4: Nurse(Id, DepartmentId), FK={DepartmentId} & Mandatory

Approach
III-A



R13: Person(Id, FirstName, MiddleName, LastName, Sex, DateOfBirth, Address, Phone)

R14: Staff(Id , PersonId)

FK={PersonId}, CK={PersonId} & Mandatory

R1: Patient(Id, Number, HealthCardNo, InsuranceName, PersonId)

CK₁={Number} & Mandatory

FK={PersonId}, CK₂={PersonId} & Mandatory

R2: Doctor(Id, MINC, SpecialtyId, StaffId),

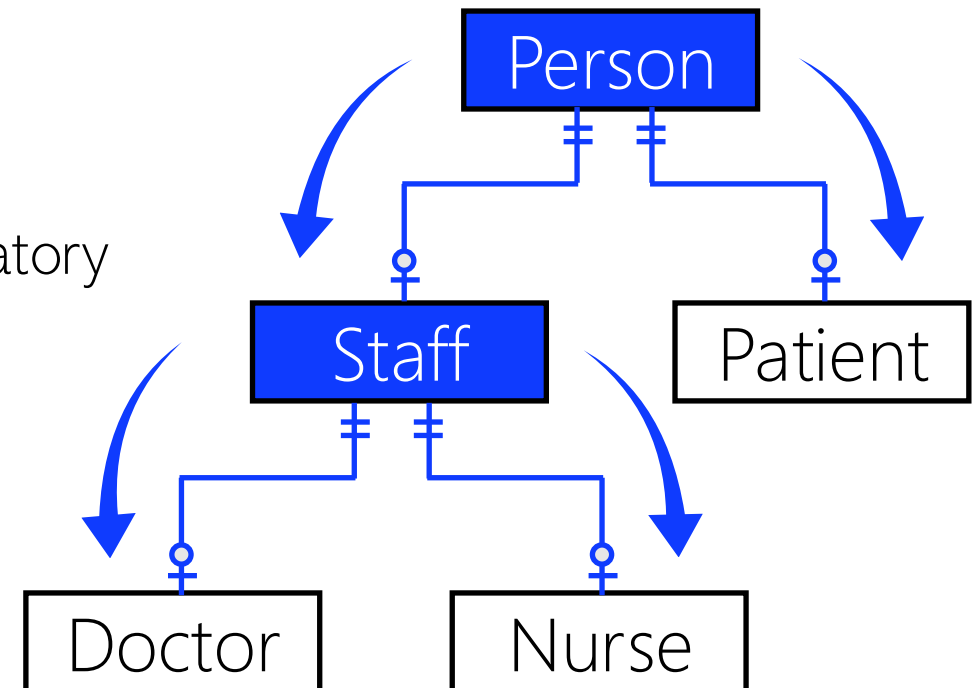
CK₁={MINC} & Mandatory, FK₁={SpecialtyId} & Mandatory

FK₂={StaffId}, CK₂={StaffId} & Mandatory

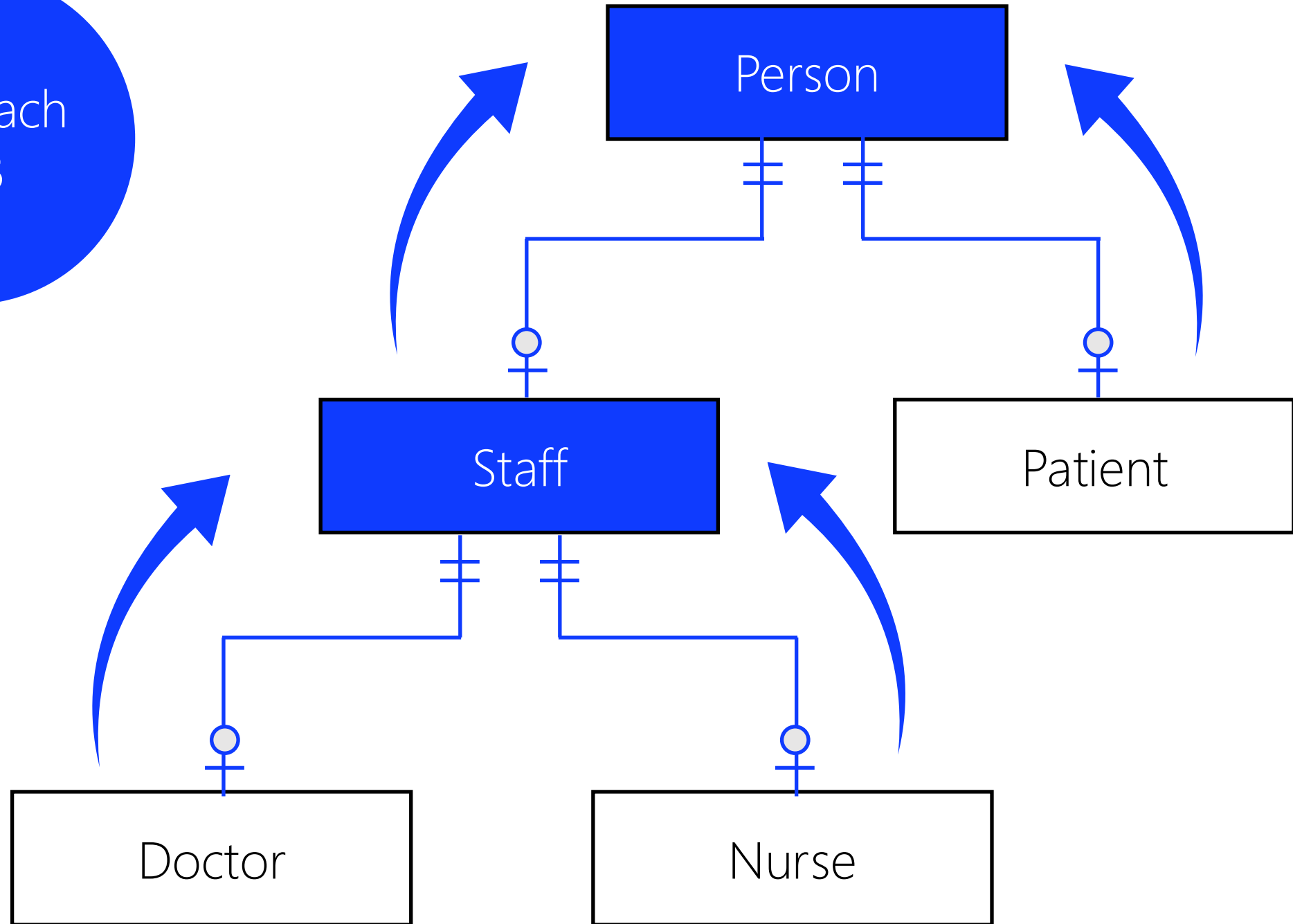
R4: Nurse(Id, DepartmentId, StaffId)

FK₁={DepartmentId} & Mandatory

FK₂={StaffId}, CK={StaffId} & Mandatory



Approach
III-B



R13: Person(Id, FirstName, MiddleName, LastName, Sex, DateOfBirth,
Address,
Phone,
StaffId,
PatientId)

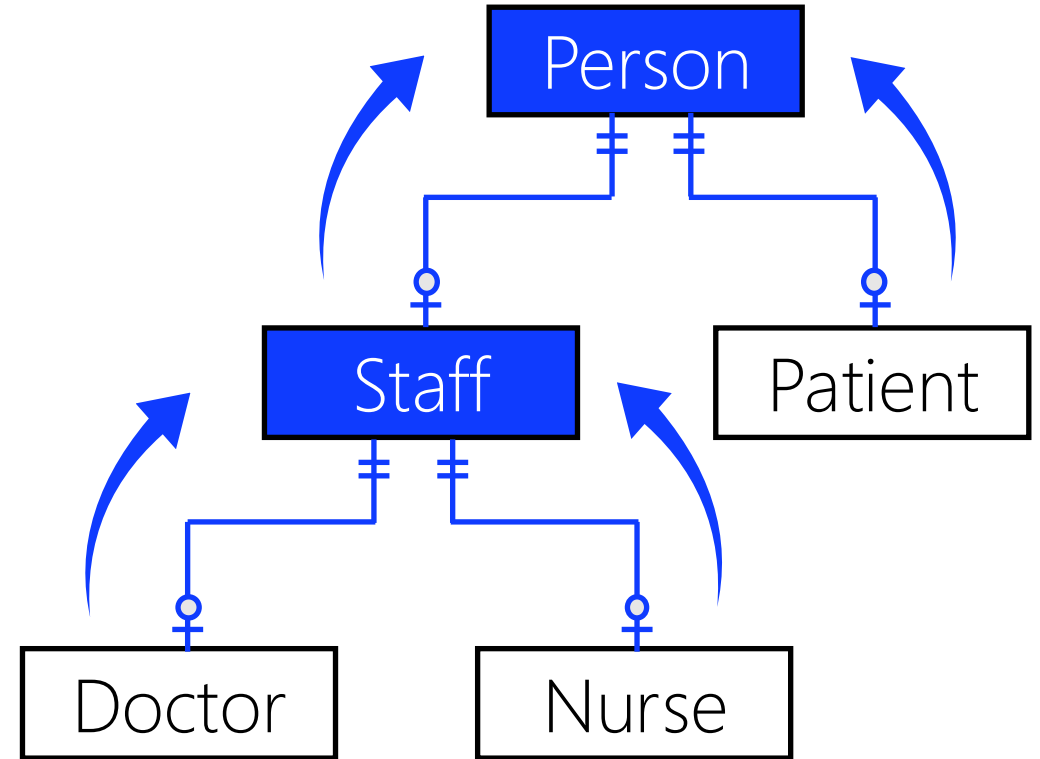
$FK_1 = \{StaffId\}$, $CK_1 = \{StaffId\}$

$FK_2 = \{PatientId\}$, $CK_2 = \{PatientId\}$

R14: Staff(Id, DoctorId, Nurseld)

$FK_1 = \{DoctorId\}$, $CK_1 = \{DoctorId\}$

$FK_2 = \{Nurseld\}$, $CK_2 = \{Nurseld\}$



Must be optional.



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